Learning a Codebase

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2 Repository: 14 **53**
- 2.1 Time series sustained components touched by developer - 14 For developer commits and components touched. . . . 53
- 2.2 Time series sustained components touched by developer - 14 For developer day and components touched. . . . 55

3 Repository: 21 **58**
- 3.1 Time series sustained components touched by developer - 21 For developer commits and components touched. . . . 58
- 3.2 Time series sustained components touched by developer - 21 For developer day and components touched. . . . 60

1 Repository

1.1 Glossary - Summary

- The founder developer starts in the first six months of a project.
- Later joiner developers start after six months.
- Transient developers have ten (10) or fewer commits.
- Moderate developers have more than ten (10) commits but fewer than 50 commits or commits for less than 250 days.
- Sustained developers make 50 or more commits and commit for 250 days or more.

1.2 Repository Summary Table

Table 1: Summary of fifteen open-source repositories identified from GitHub that have at least 1000 pull requests and zero or more sustained late joiner developers. Sustained later joiner developers who joined the project after six months and contributed at least 50 commits over a period of 250 days or more. Please note that five developers each worked on two repositories.

ID	Repo Name	Transient Founder	Moderate Founder	Sustained Founder	Transient Later Joiner	Moderate Later Joiner	Sustained Later Joiner	Commit	Start	End
3	activiti	6	7	6	212	20	5	8662	2010-Jun-18	Ongoing
5	airbyte-platform	16	2	1	457	32	6	6784	2020-Aug-04	2023-Sep-27
7	ambari	0	3	0	129	32	15	4247	2011-Sep-22	Ongoing
10	automq	4	0	0	301	10	3	1874	2011-Aug-10	2017-Apr-21
14	buck	18	5	6	531	85	21	14876	2013-Mar-21	2022-May-05
15	camel	1	4	1	1146	63	13	36528	2007-Mar-19	Ongoing
18	checkstyle	0	0	1	362	42	9	17338	2001-Jun-22	Ongoing
20	cxf	6	6	3	201	17	3	7893	2008-Apr-29	Ongoing
21	intellij-community	0	3	7	344	93	84	101247	2004-Nov-11	2018-Apr-18
26	guava	0	1	0	402	8	3	9685	2009-Sep-15	Ongoing
28	jenkins	5	1	1	837	39	4	30252	2006-Nov-05	Ongoing
33	openmrs-core	0	1	0	388	26	3	15757	2006-May-03	Ongoing
36	presto	0	1	3	692	71	18	30605	2012-Aug-09	Ongoing
37	quarkus	2	3	5	824	25	8	81612	2018-Jun-14	Ongoing
39	selenium	5	3	0	712	17	8	23354	2004-Nov-03	Ongoing
	Total	63	40	34	7538	580	203	390714		

1.3 Sample scatter touched against commit -

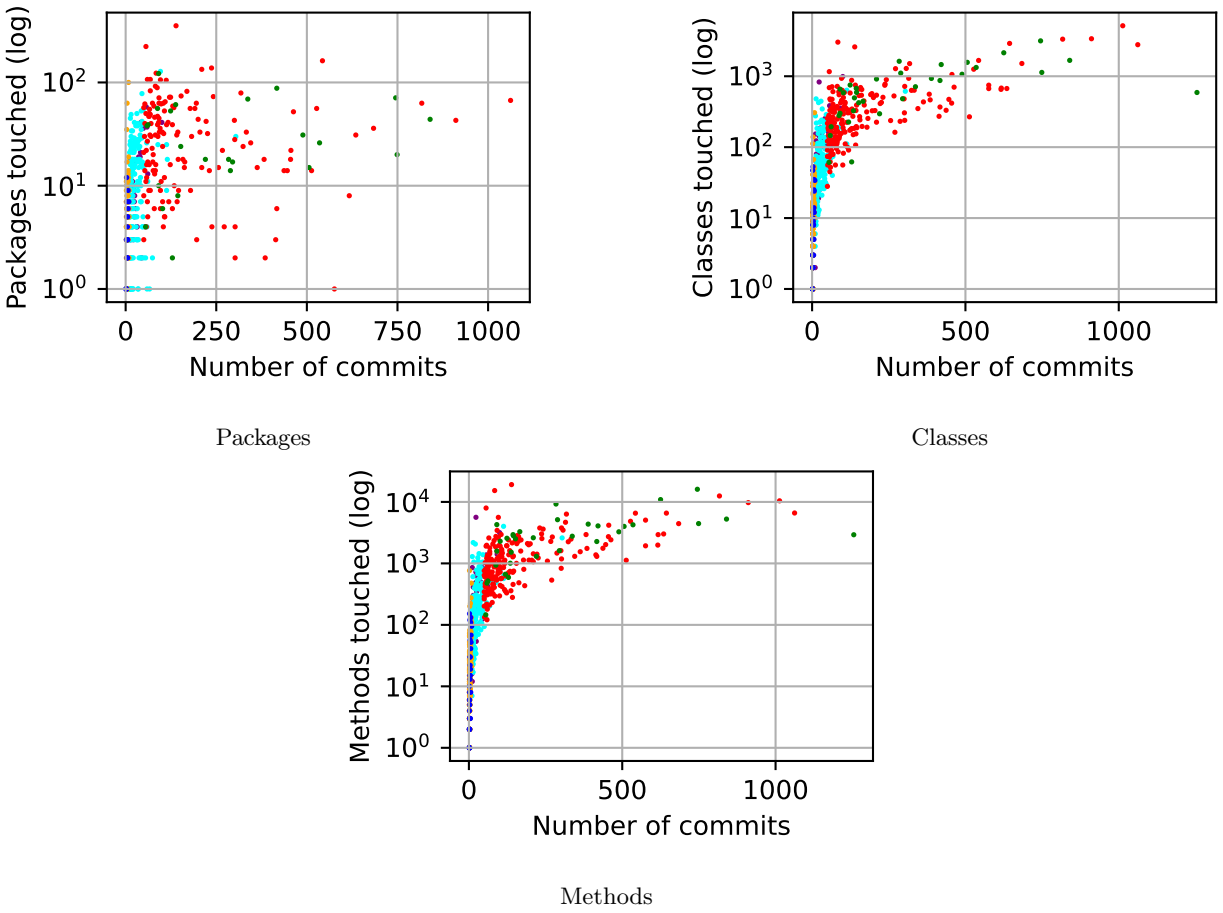
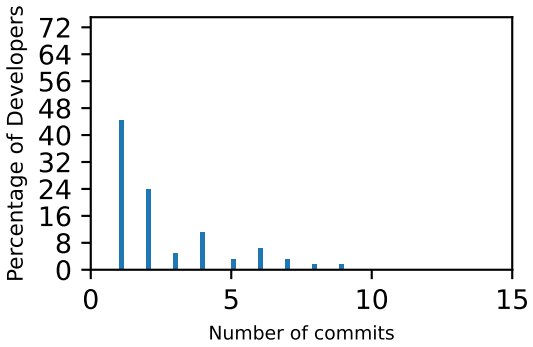
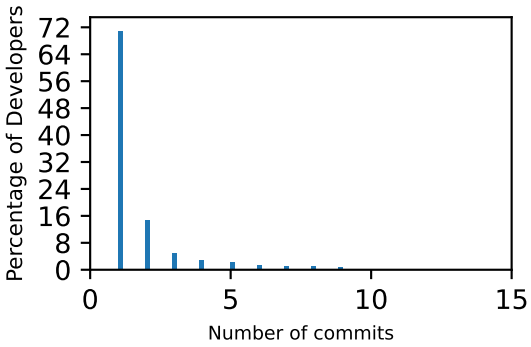


Figure 1: Scatter plots of the log total number of components touched (y-axis) against the number of commits (x-axis) made by samples of developers capped at 203 sustained later joiners from six (6) categories: Transient Founder (Blue, 63), Moderate Founder (Purple, 40), Sustained Founder (Green, 34), Transient Later Joiner (Orange, 203), Moderate Later Joiner (Cyan, 203), Sustained Later Joiner (Red, 203).

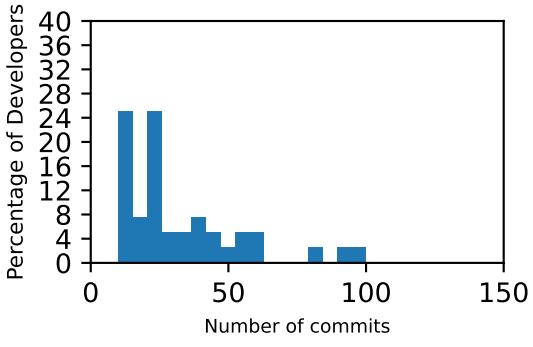
1.4 Total commits by developer - Developer Commits



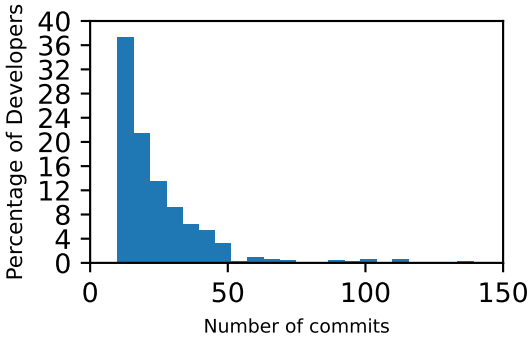
Transient founder (n=63).



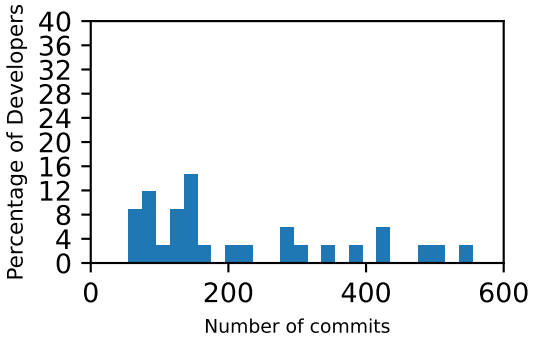
Transient later joiner (n=7520).



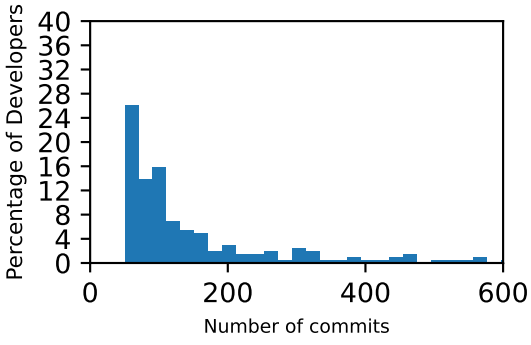
Moderate founder (n=40).



Moderate later joiner (n=580).



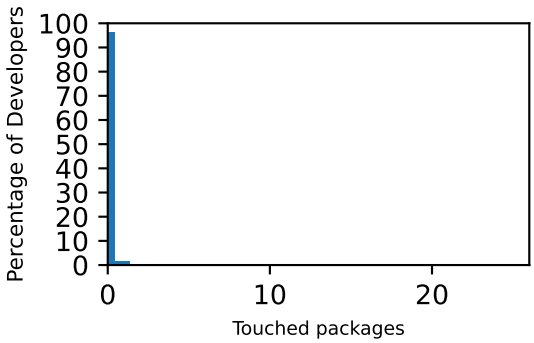
Sustained founder (n=34).



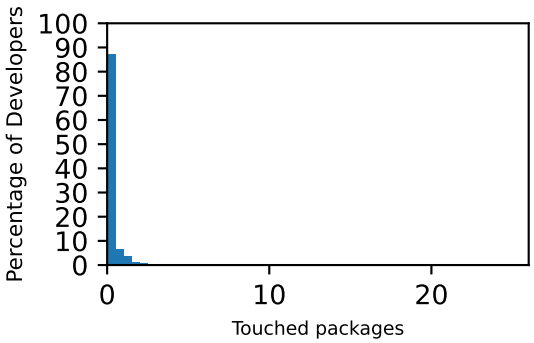
Sustained later joiner (n=203).

Figure 2: Histogram of percentage of developers (y-axis) against total commits made (x-axis) in six (6) categories. This is from 15 repositories sampled from GitHub.

1.5 Histogram components touched by developer - For components with total

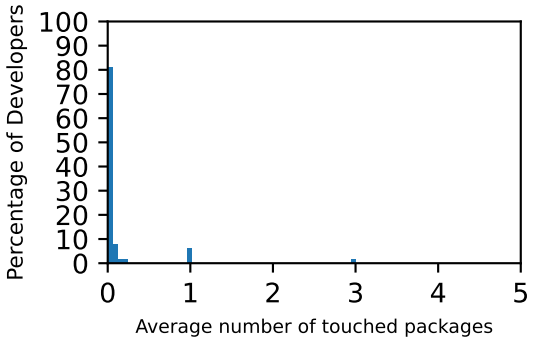


By Day.

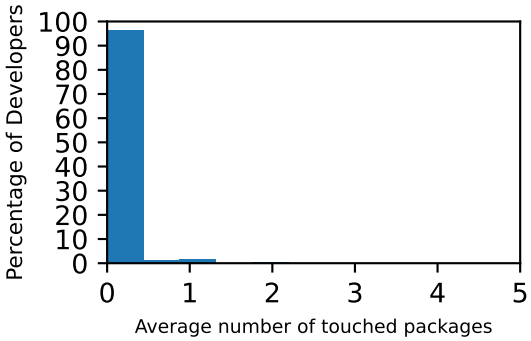


By Commit.

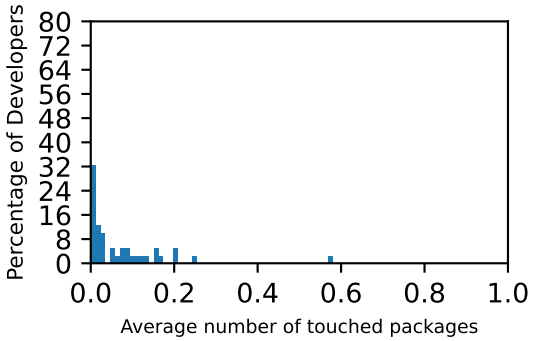
Figure 3: Histogram of percentage of all developers (y-axis) against the average packages touched by day and commit (x-axis) from all 8426 developers from 15 repositories sampled from GitHub.



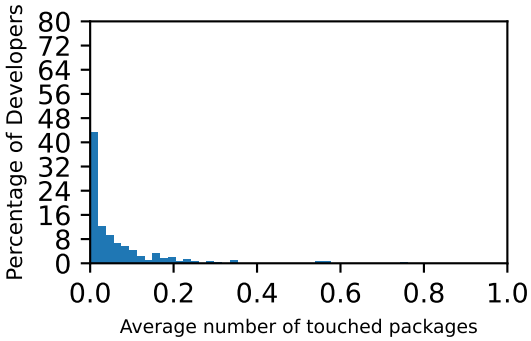
Transient founder (n=63).



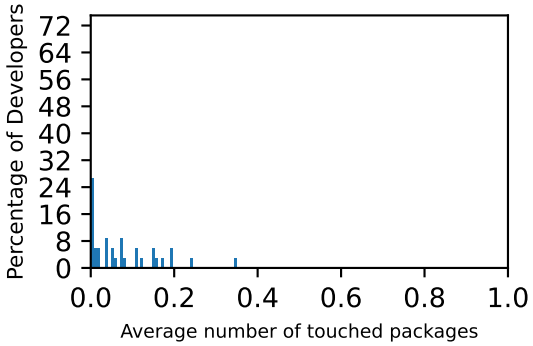
Transient later joiner (n=7520).



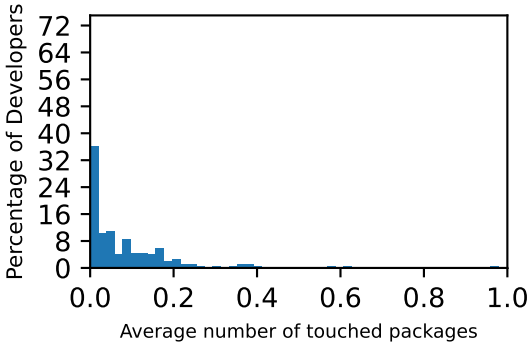
Moderate founder (n=40).



Moderate later joiner (n=580).

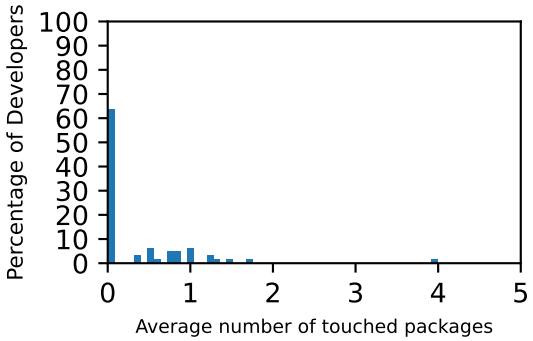


Sustained founder (n=34).

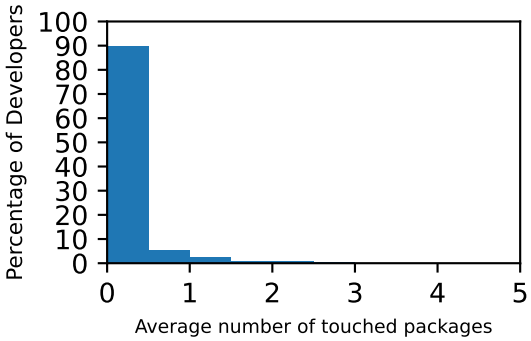


Sustained later joiner (n=203).

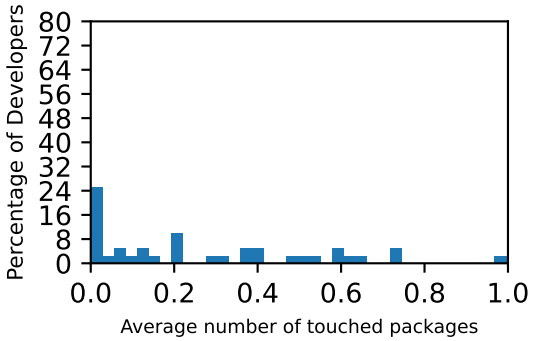
Figure 4: Histogram of percentage of developers (y-axis) against the average packages touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



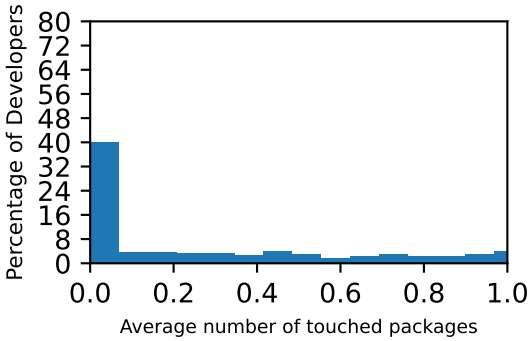
Transient founder (n=63).



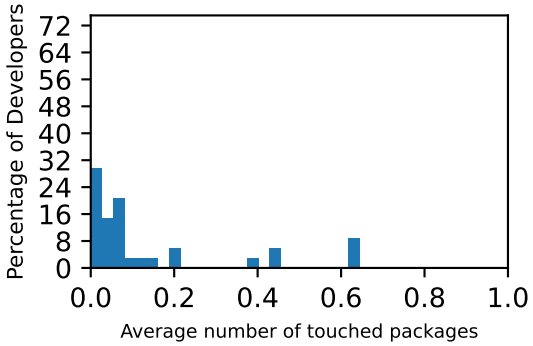
Transient later joiner (n=7520).



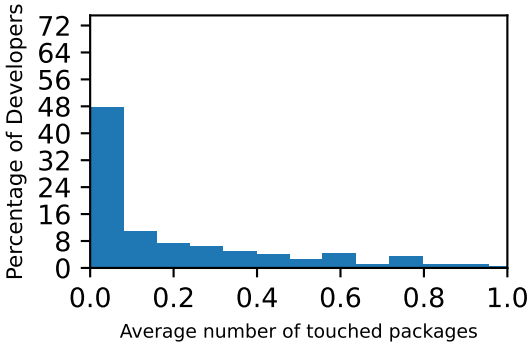
Moderate founder (n=40).



Moderate later joiner (n=580).

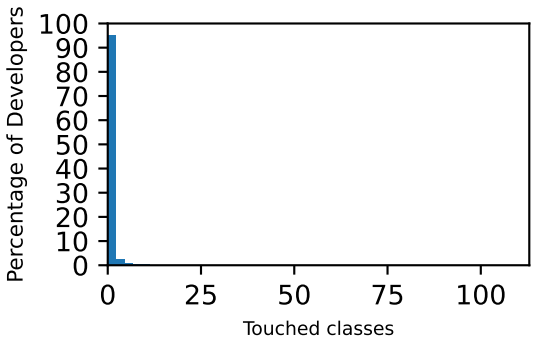


Sustained founder (n=34).

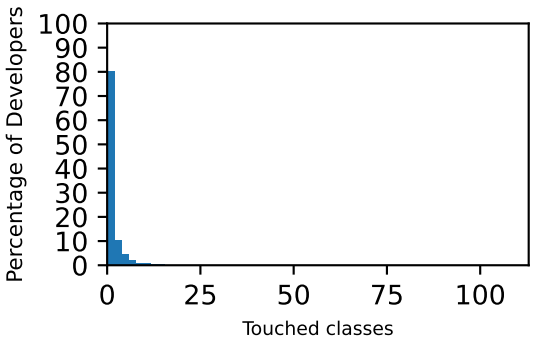


Sustained later joiner (n=203).

Figure 5: Histogram of percentage of developers (y-axis) against the average packages touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.

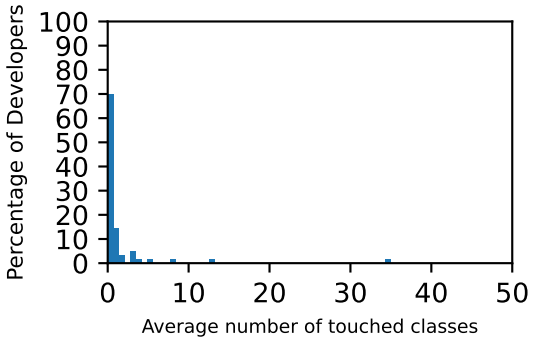


By Day.

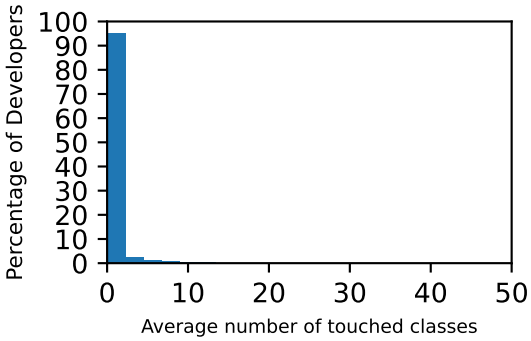


By Commit.

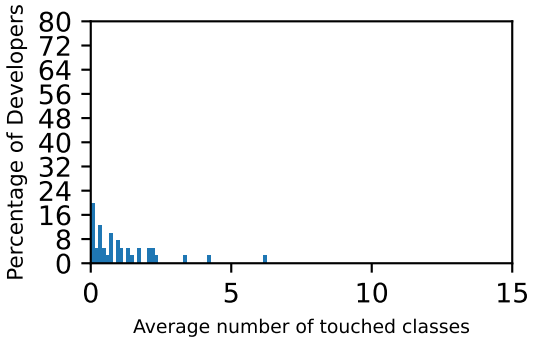
Figure 6: Histogram of percentage of all developers (y-axis) against the average classes touched by day and commit (x-axis) from all 8426 developers from 15 repositories sampled from GitHub.



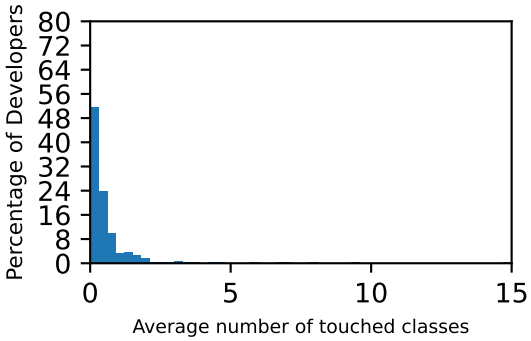
Transient founder (n=63).



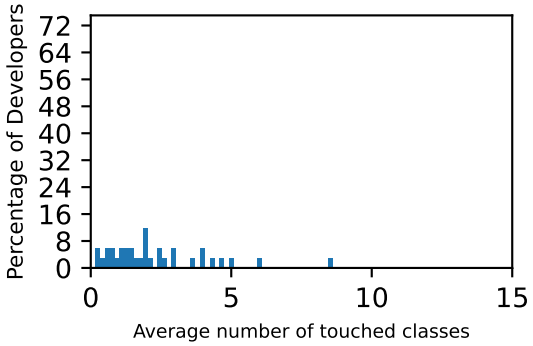
Transient later joiner (n=7520).



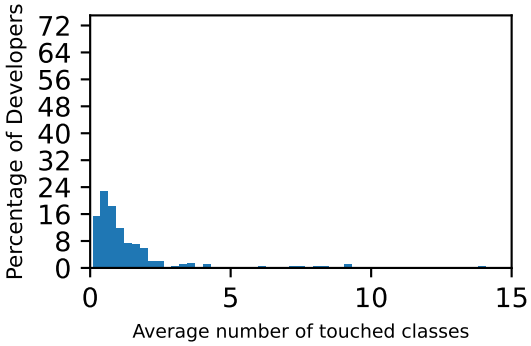
Moderate founder (n=40).



Moderate later joiner (n=580).

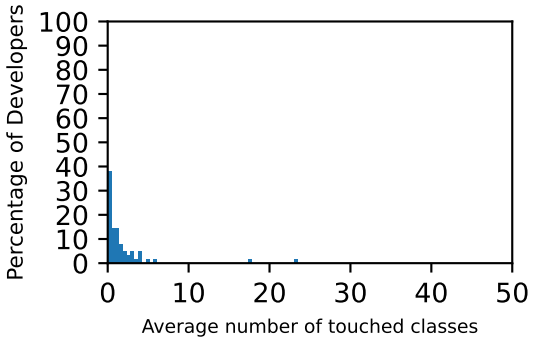


Sustained founder (n=34).

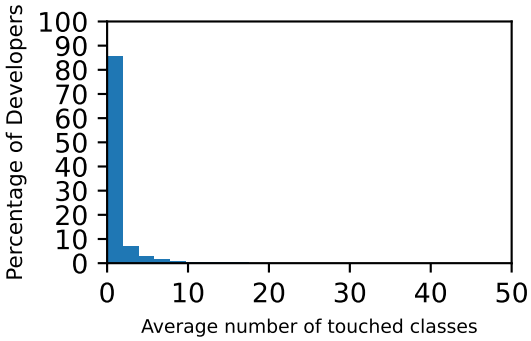


Sustained later joiner (n=203).

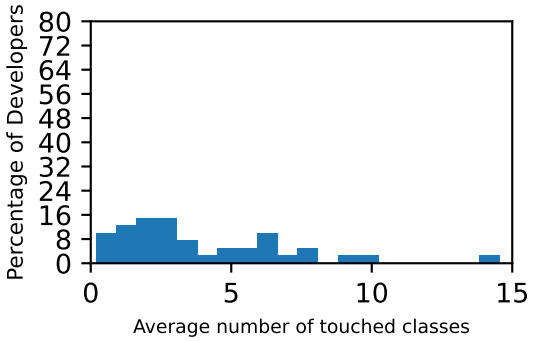
Figure 7: Histogram of percentage of developers (y-axis) against the average classes touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



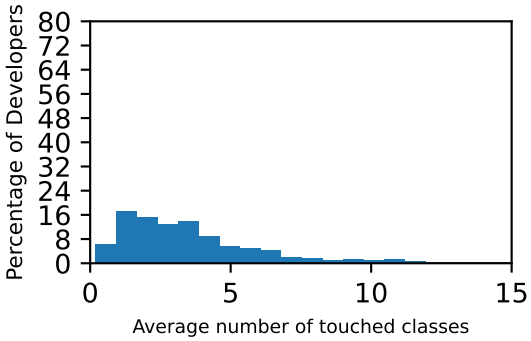
Transient founder (n=63).



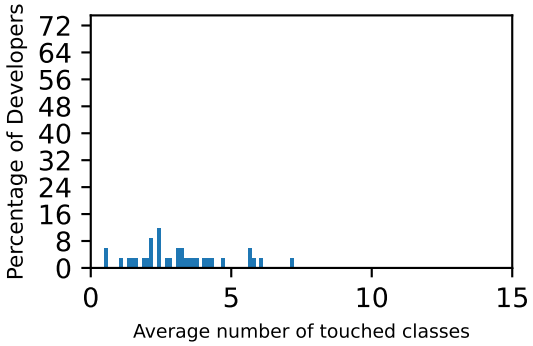
Transient later joiner (n=7520).



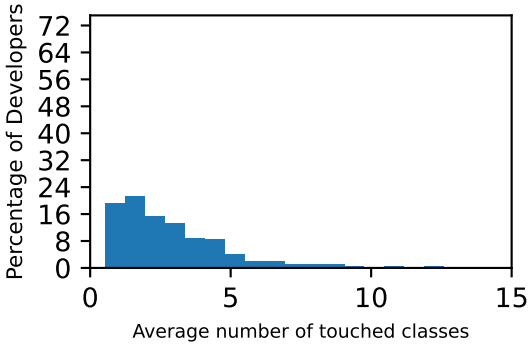
Moderate founder (n=40).



Moderate later joiner (n=580).

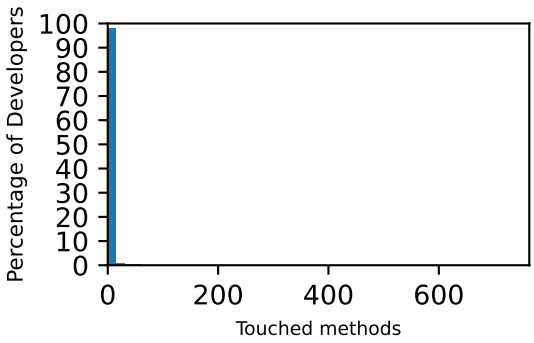


Sustained founder (n=34).

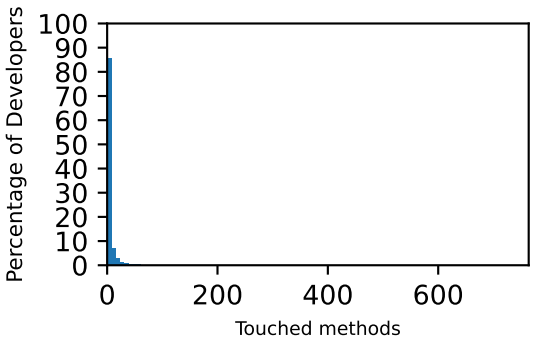


Sustained later joiner (n=203).

Figure 8: Histogram of percentage of developers (y-axis) against the average classes touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.

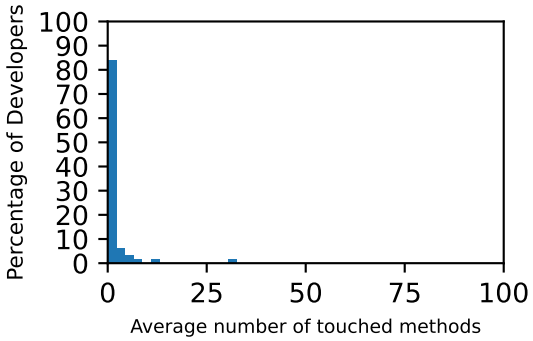


By Day.

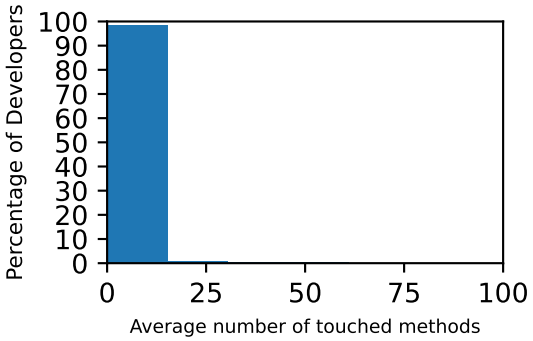


By Commit.

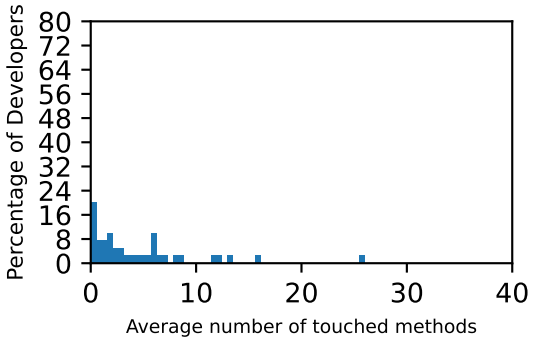
Figure 9: Histogram of percentage of all developers (y-axis) against the average methods touched by day and commit (x-axis) from all 8426 developers from 15 repositories sampled from GitHub.



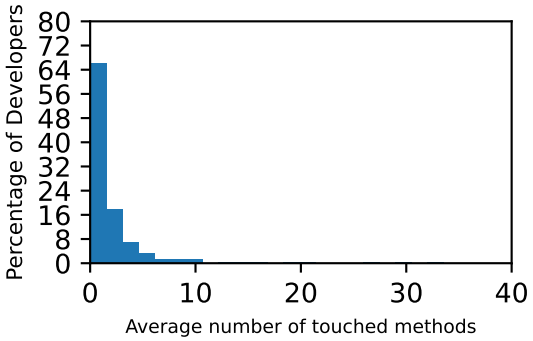
Transient founder (n=63).



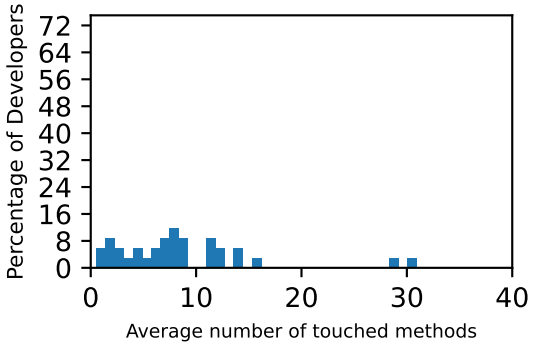
Transient later joiner (n=7520).



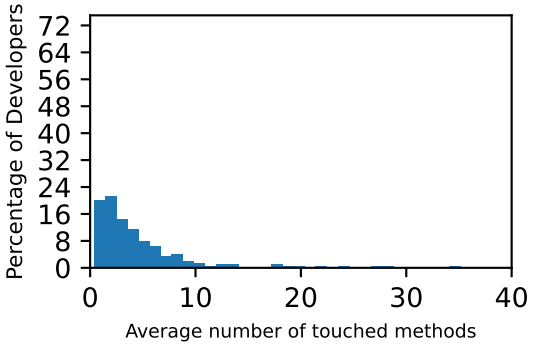
Moderate founder (n=40).



Moderate later joiner (n=580).

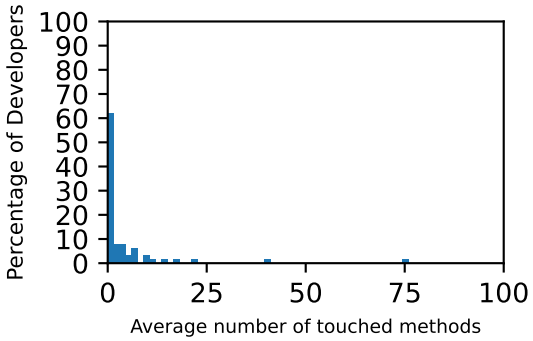


Sustained founder (n=34).

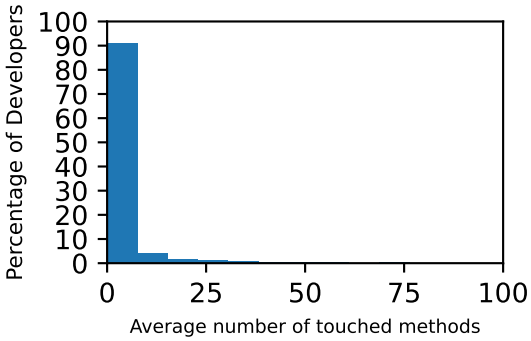


Sustained later joiner (n=203).

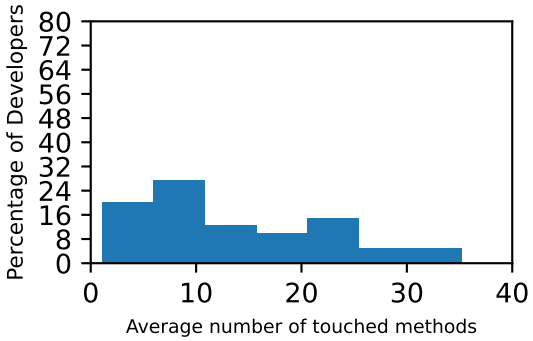
Figure 10: Histogram of percentage of developers (y-axis) against the average methods touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



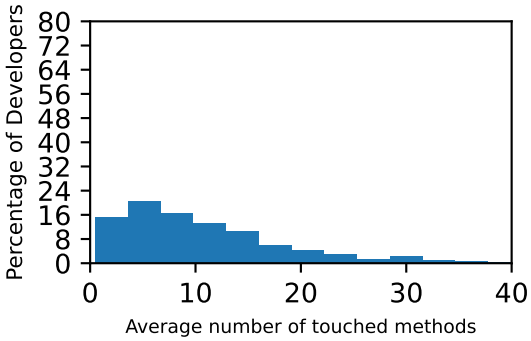
Transient founder (n=63).



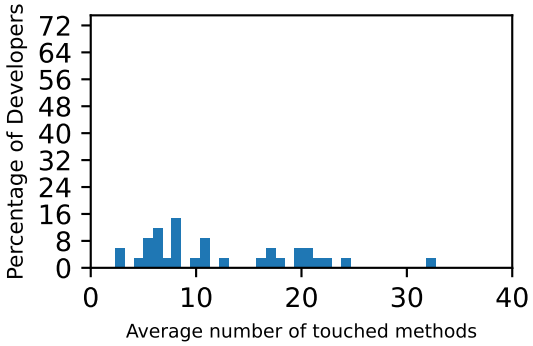
Transient later joiner (n=7520).



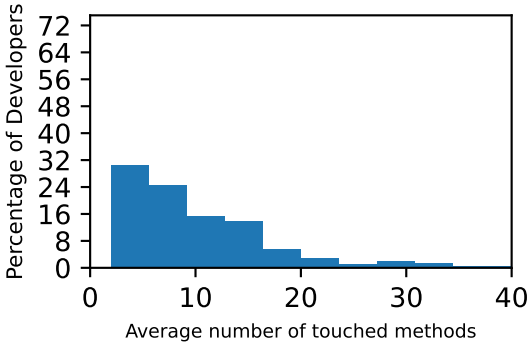
Moderate founder (n=40).



Moderate later joiner (n=580).



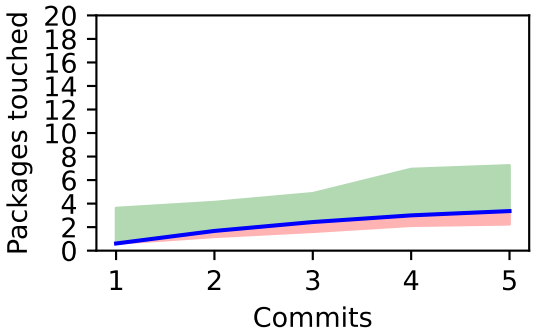
Sustained founder (n=34).



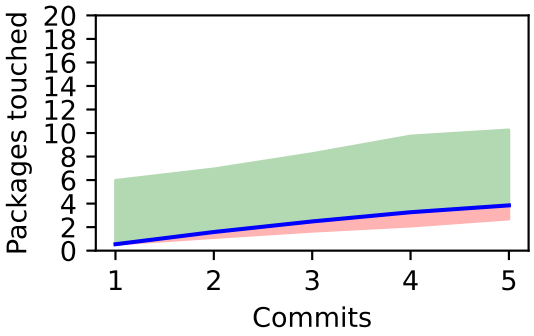
Sustained later joiner (n=203).

Figure 11: Histogram of percentage of developers (y-axis) against the average methods touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.

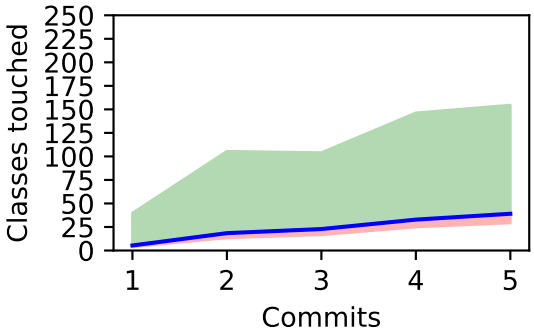
1.5.1 Time series components touched - For packages touched for first five commits



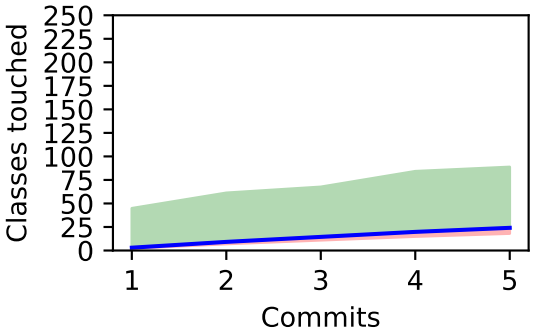
Packages for All founders (n=74 & μ=3.36)



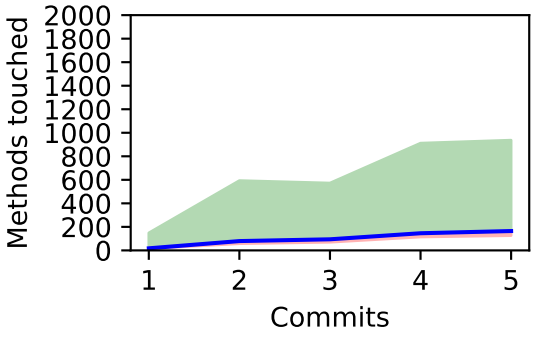
Packages for All joiners (n=783 & μ=3.85)



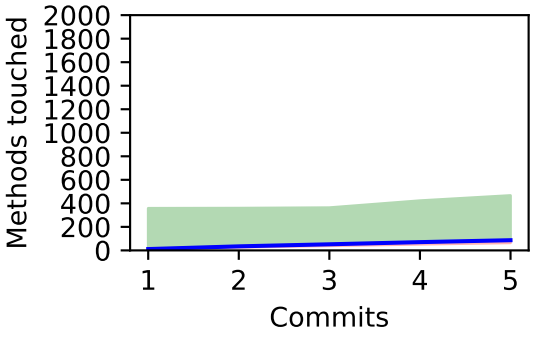
Classes for All founders (n=74 & μ=39.05)



Classes for All joiners (n=783 & μ=24.11)

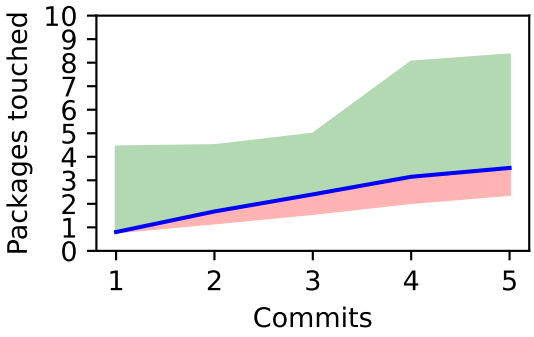


Methods for All founders (n=74 & μ=164.59)

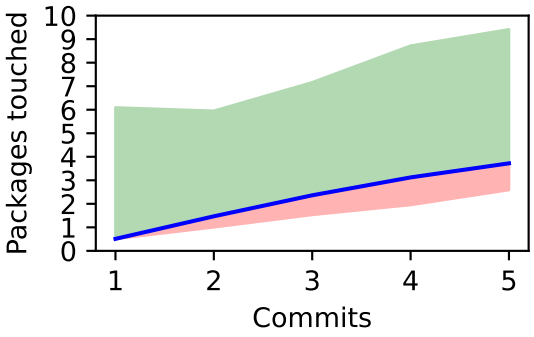


Methods for All joiners (n=783 & μ=86.68)

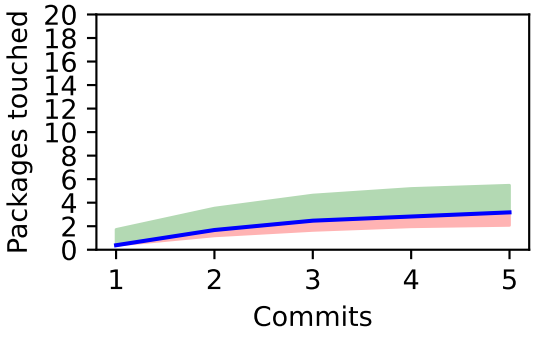
Figure 12: A time series for the five (5) commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



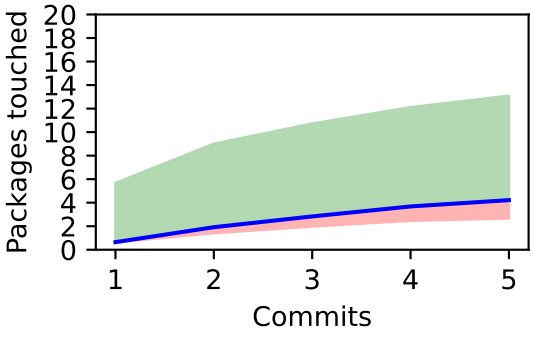
Moderate founder (n=40 & μ=3.52)



Moderate later joiner (n=580 & μ=3.72)

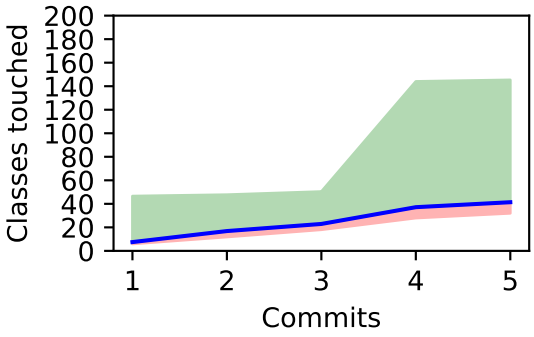


Sustained founder (n=34 & μ=3.18)

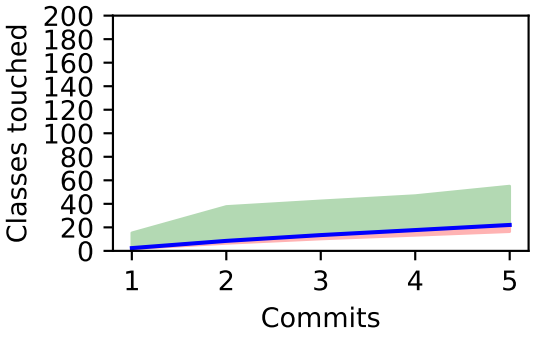


Sustained later joiner (n=203 & μ=4.22)

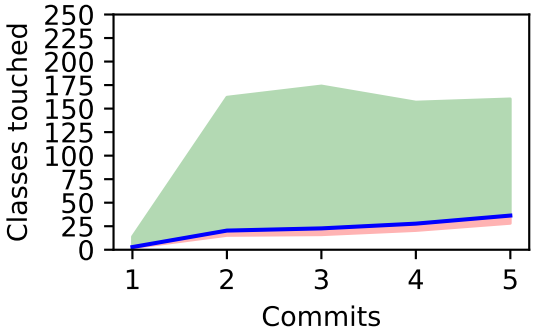
Figure 13: A time series for the five (5) commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



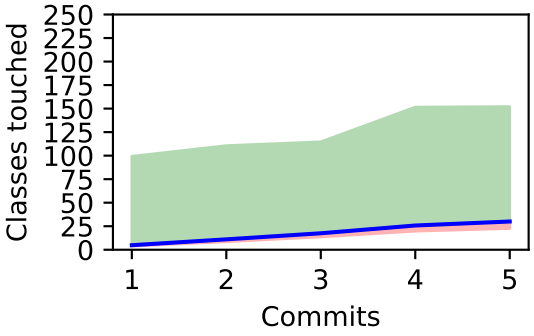
Moderate founder (n=40 & μ=41.38)



Moderate later joiner (n=580 & μ=22.04)

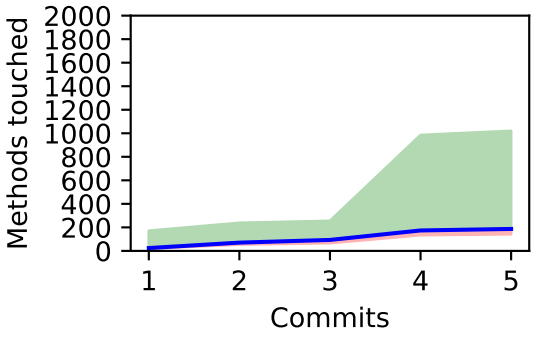


Sustained founder (n=34 & μ=36.32)

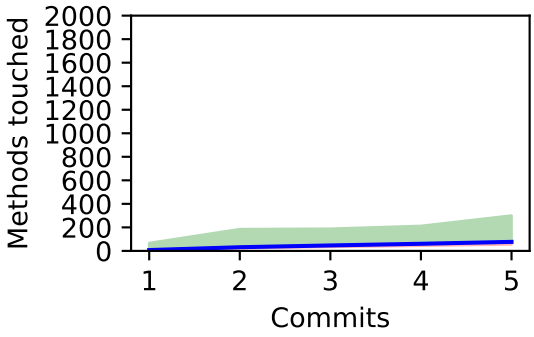


Sustained later joiner (n=203 & μ=30.01)

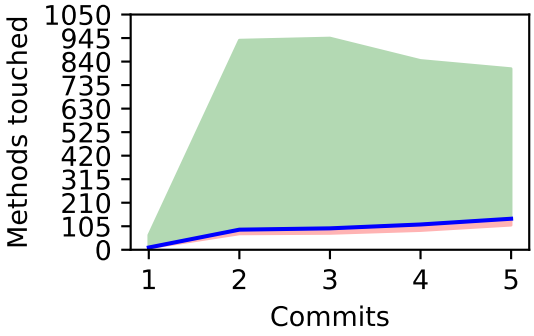
Figure 14: A time series for the five (5) commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



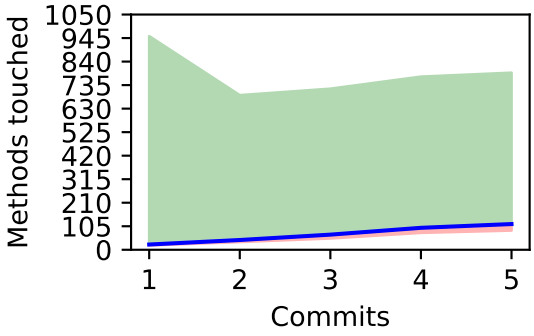
Moderate founder (n=40 & μ=186.75)



Moderate later joiner (n=580 & μ=76.85)



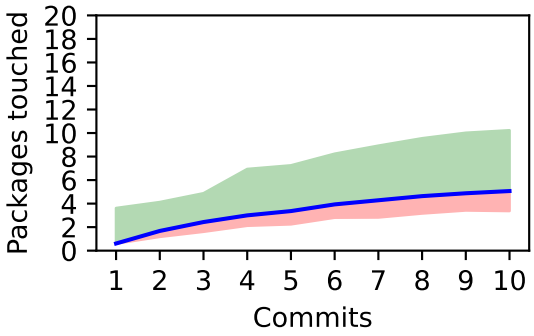
Sustained founder (n=34 & μ=138.53)



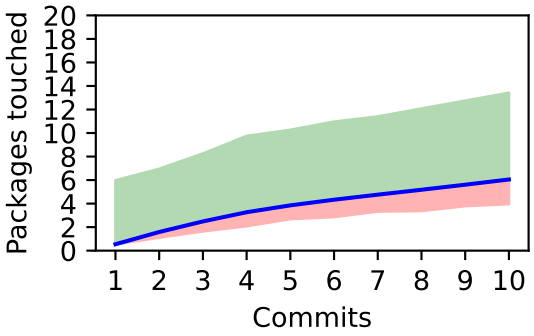
Sustained later joiner (n=203 & μ=114.77)

Figure 15: A time series for the five (5) commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

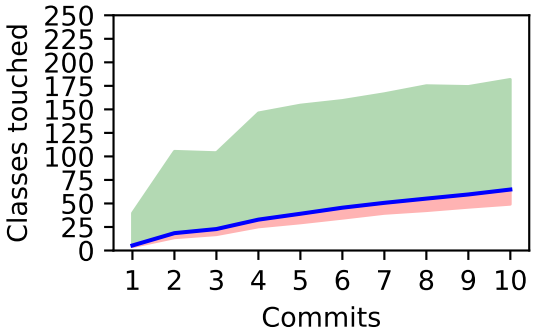
1.5.2 Time series components touched - For packages touched for first ten commits



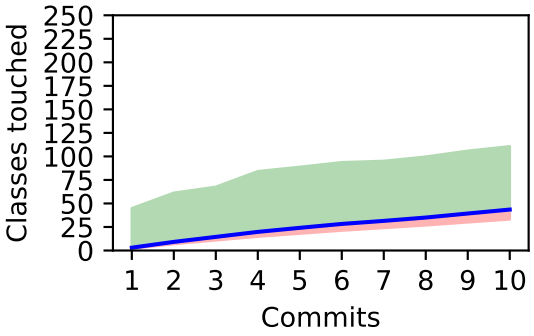
Packages for All founders (n=74 & μ=5.07)



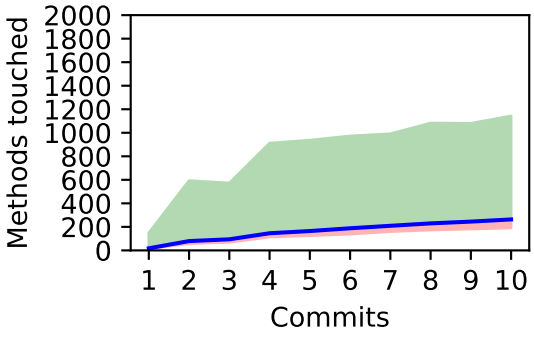
Packages for All joiners (n=783 & μ=6.05)



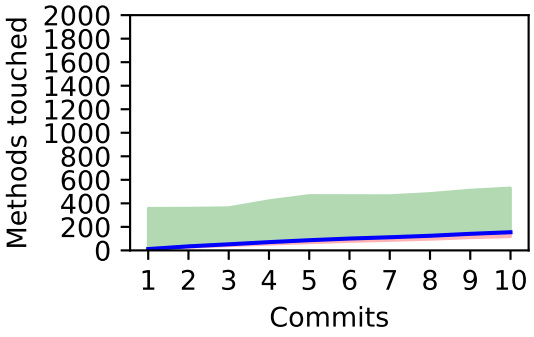
Classes for All founders (n=74 & μ=64.81)



Classes for All joiners (n=783 & μ=43.51)

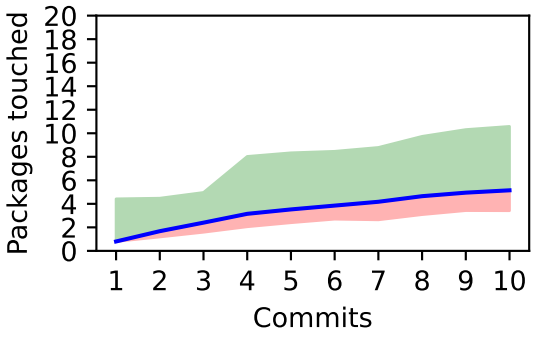


Methods for All founders (n=74 & μ=263.26)

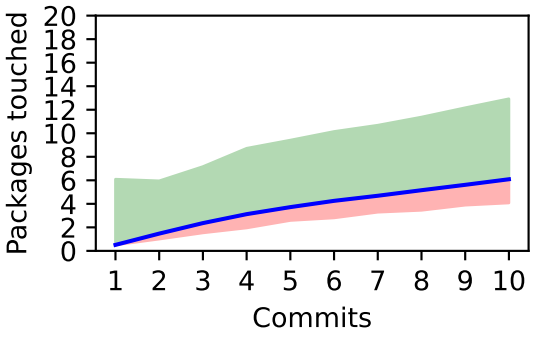


Methods for All joiners (n=783 & μ=154.41)

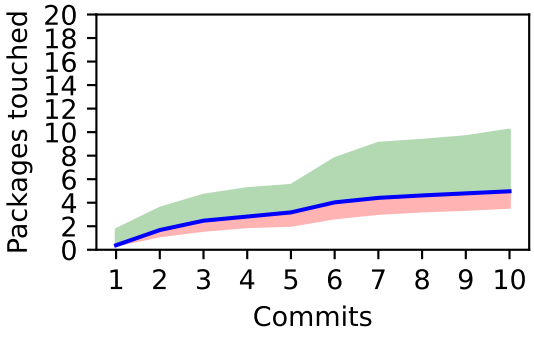
Figure 16: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



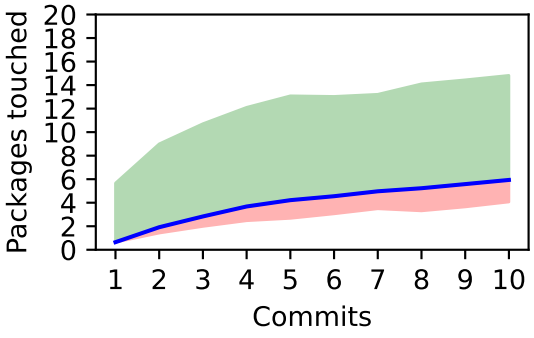
Moderate founder (n=40 & μ=5.15)



Moderate later joiner (n=580 & μ=6.09)

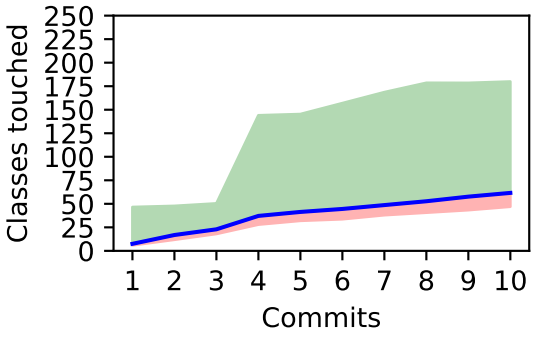


Sustained founder (n=34 & μ=4.97)

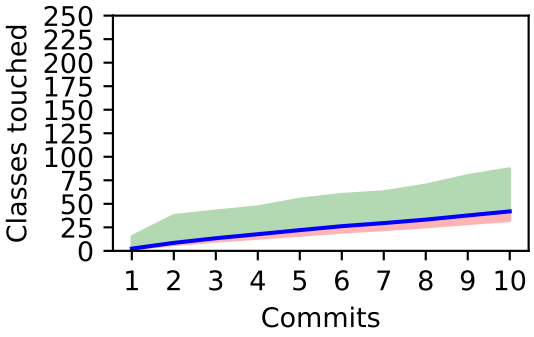


Sustained later joiner (n=203 & μ=5.94)

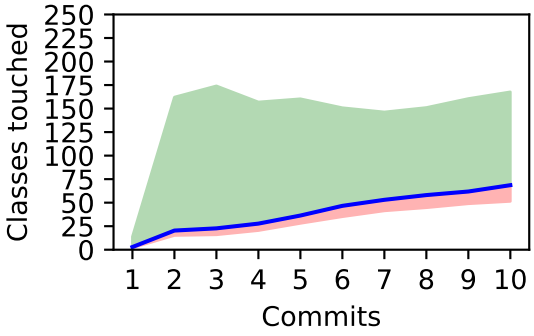
Figure 17: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



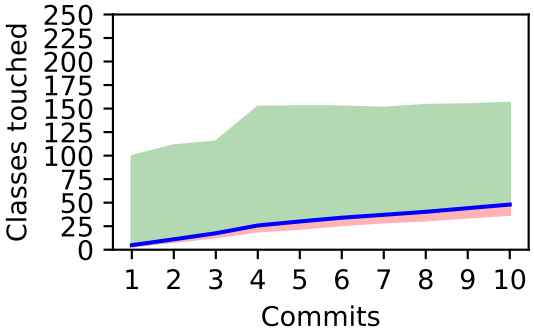
Moderate founder (n=40 & μ=61.6)



Moderate later joiner (n=580 & μ=41.95)

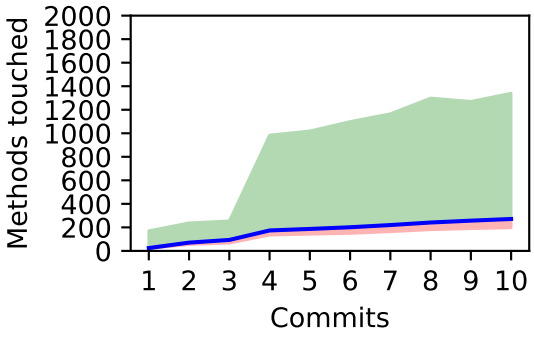


Sustained founder (n=34 & μ=68.59)

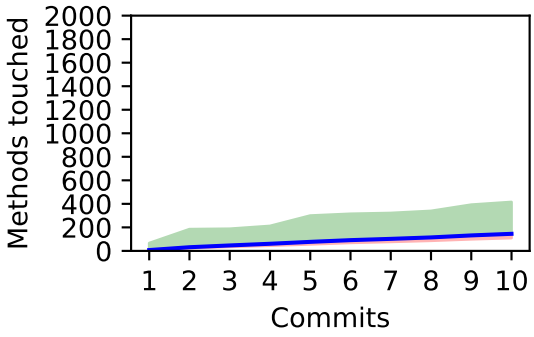


Sustained later joiner (n=203 & μ=47.98)

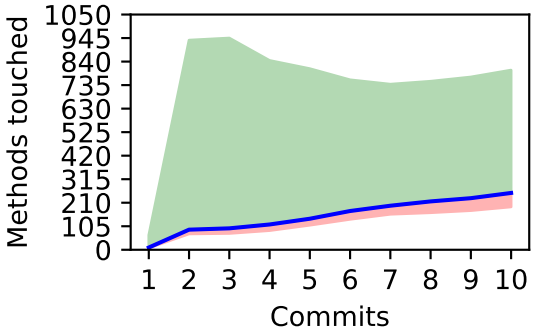
Figure 18: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



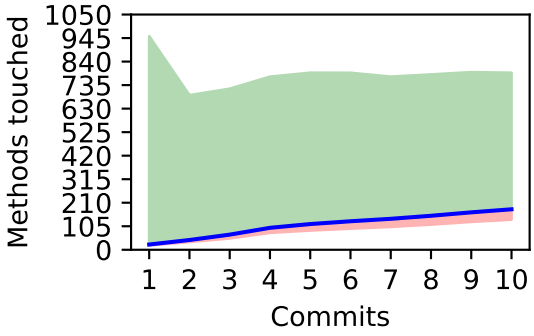
Moderate founder (n=40 & μ=271.42)



Moderate later joiner (n=580 & μ=145.21)



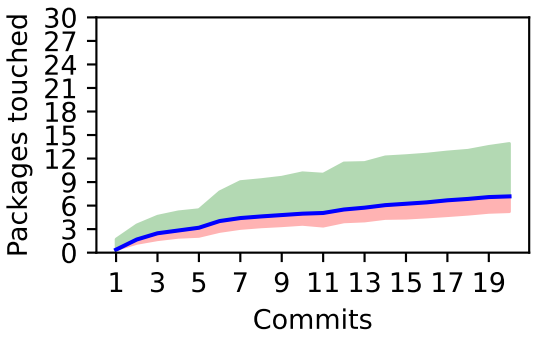
Sustained founder (n=34 & μ=253.65)



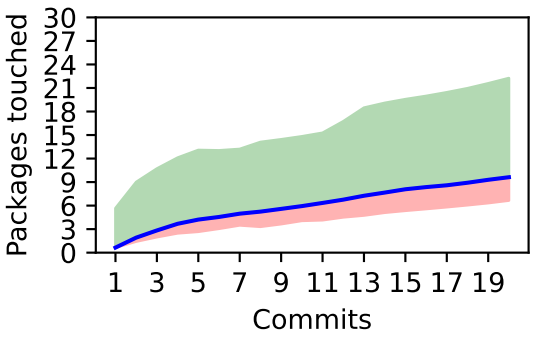
Sustained later joiner (n=203 & μ=180.7)

Figure 19: A time series for the ten (10) commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for four (4) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

1.5.3 Time series components touched - For packages touched for first twenty commits

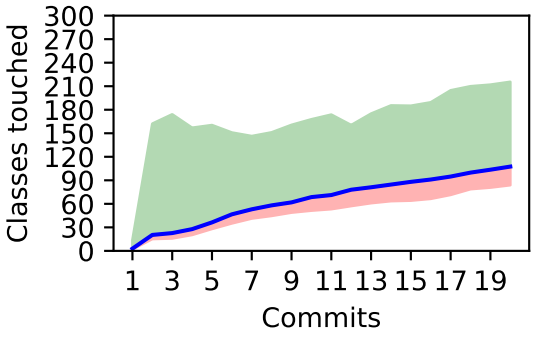


Sustained founder (n=34 & μ=7.18)

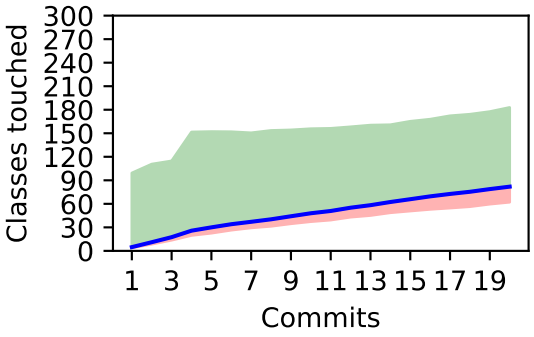


Sustained later joiner (n=203 & μ=9.62)

Figure 20: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

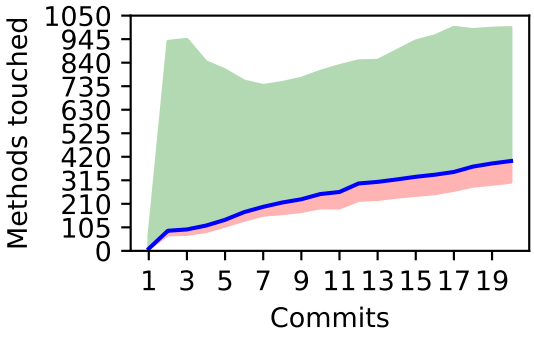


Sustained founder (n=34 & μ=107.5)

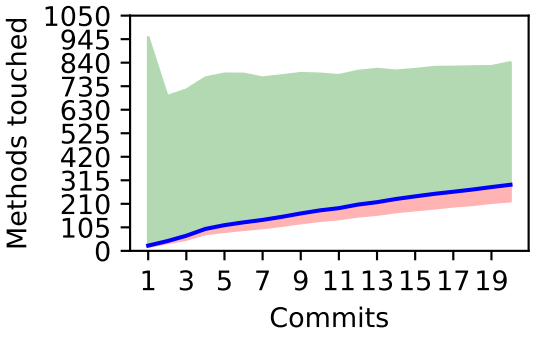


Sustained later joiner (n=203 & μ=81.91)

Figure 21: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



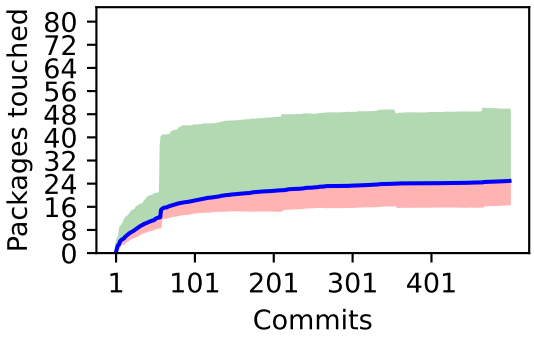
Sustained founder (n=34 & μ=401.24)



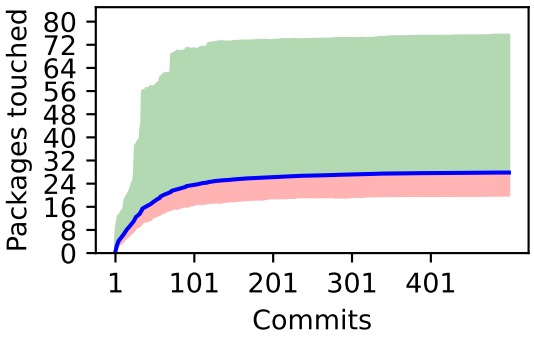
Sustained later joiner (n=203 & μ=295.1)

Figure 22: A time series for the twenty (20) commits, sustained developers, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

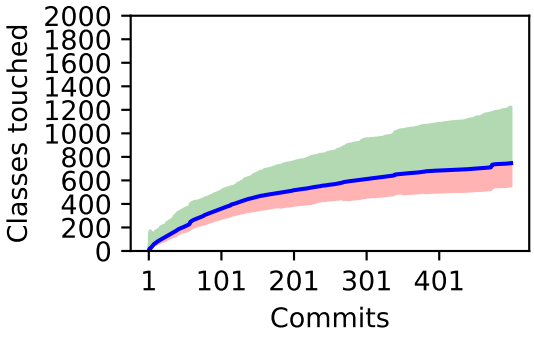
1.5.4 Time series components touched - For packages touched for commit



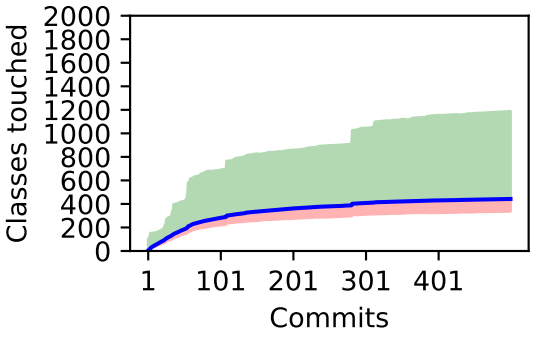
Packages for All founders (n=34 & μ=24.94)



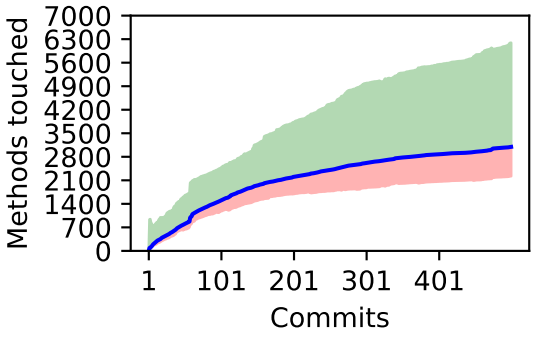
Packages for All joiners (n=203 & μ=27.84)



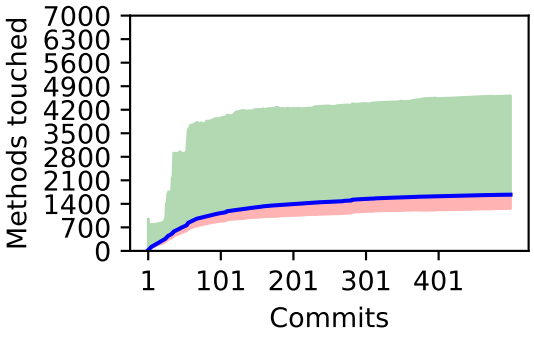
Classes for All founders (n=34 & μ=746.59)



Classes for All joiners (n=203 & μ=442.33)

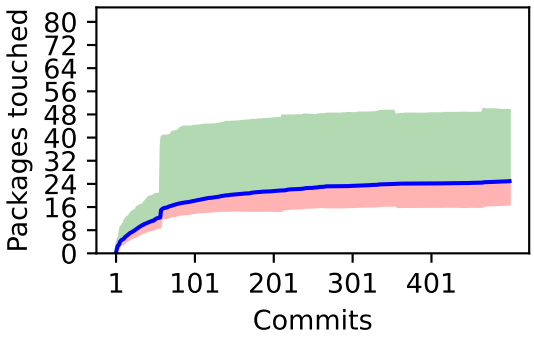


Methods for All founders (n=34 & μ=3095.18)

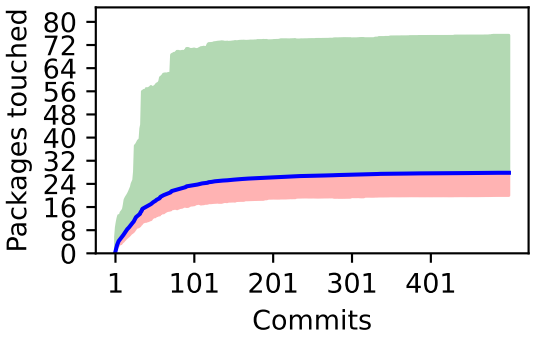


Methods for All joiners (n=203 & μ=1672.84)

Figure 23: A time series for the all commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

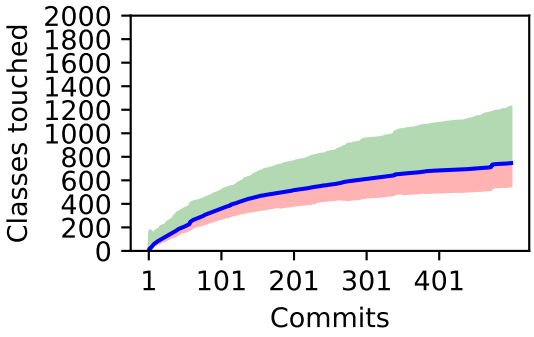


Sustained founder (n=34 & μ=24.94)

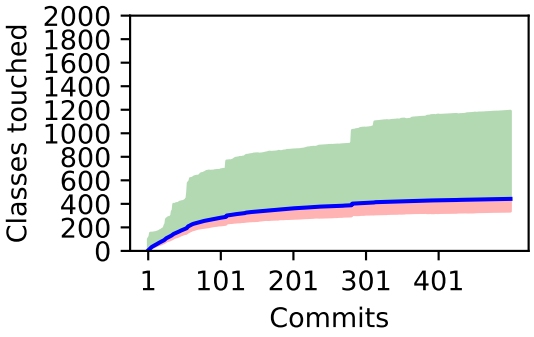


Sustained later joiner (n=203 & μ=27.84)

Figure 24: A time series for the all commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

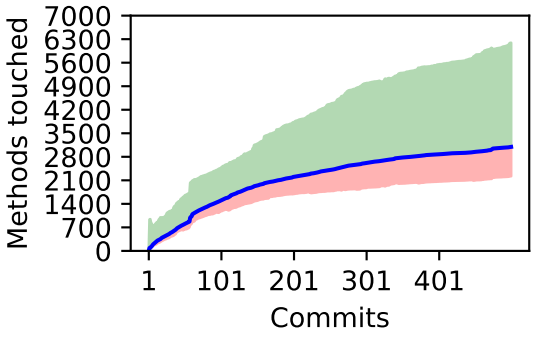


Sustained founder (n=34 & μ=746.59)

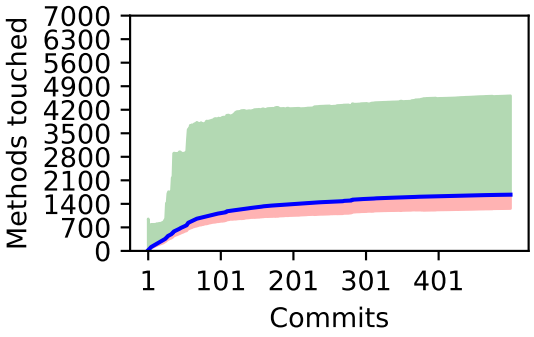


Sustained later joiner (n=203 & μ=442.33)

Figure 25: A time series for the all commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



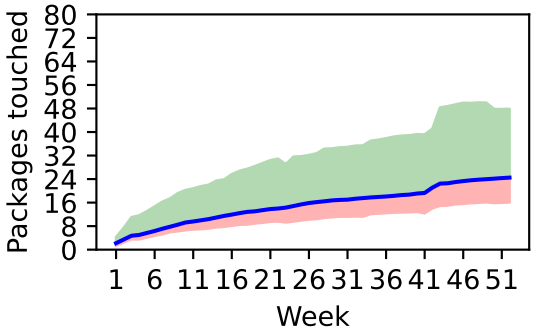
Sustained founder (n=34 & μ=3095.18)



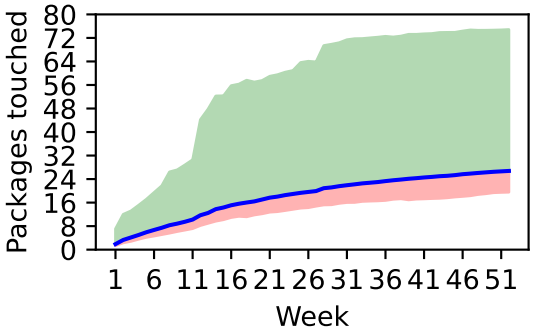
Sustained later joiner (n=203 & μ=1672.84)

Figure 26: A time series for the all commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of commits (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

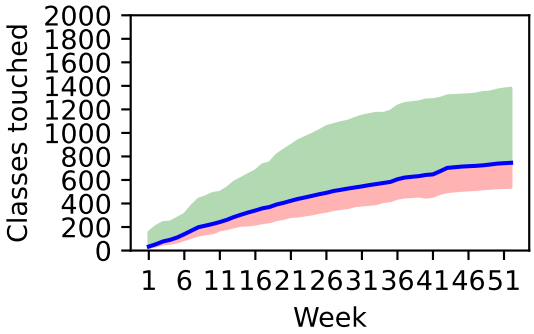
1.5.5 Time series components touched - For packages touched for each week commit



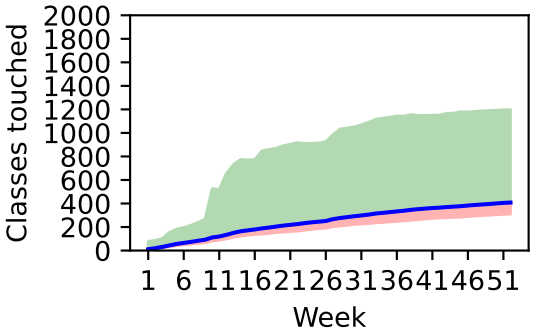
Packages for All founders (n=34 & μ=24.5)



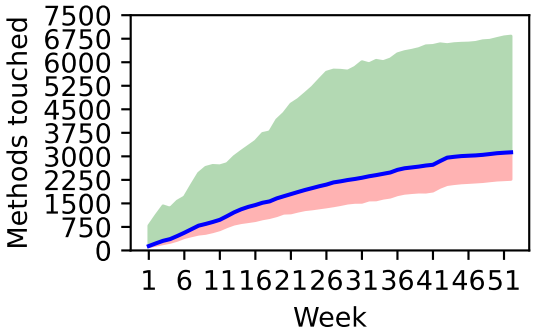
Packages for All joiners (n=203 & μ=26.77)



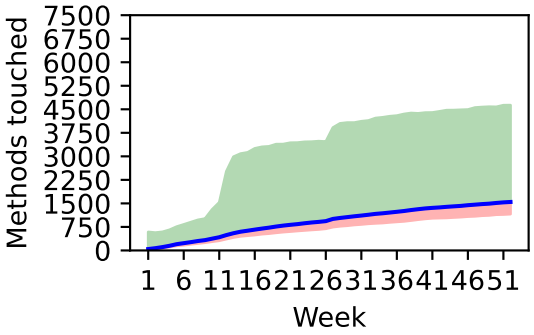
Classes for All founders (n=34 & μ=746.15)



Classes for All joiners (n=203 & μ=408.05)

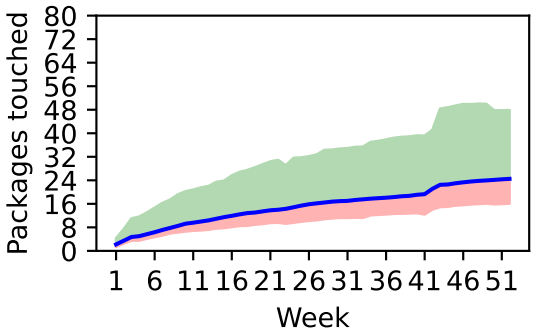


Methods for All founders (n=34 & μ=3129.0)

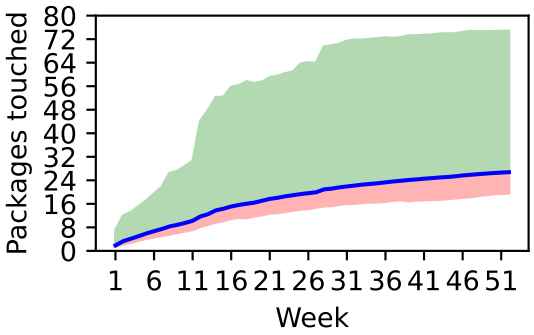


Methods for All joiners (n=203 & μ=1545.61)

Figure 27: A time series for the all commits, excluding transient, with the average mean (blue line), all components touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

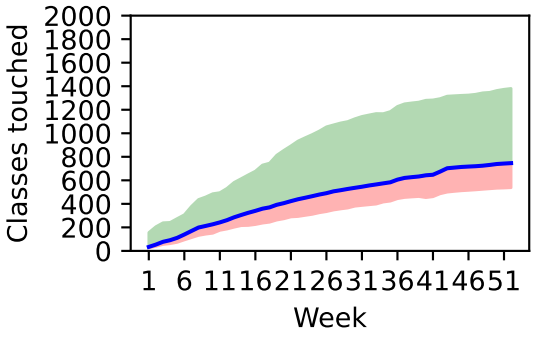


Sustained founder (n=34 & μ=24.5)

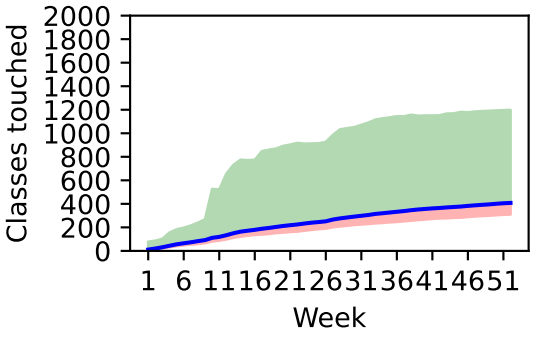


Sustained later joiner (n=203 & μ=26.77)

Figure 28: A time series for the all commits, excluding transient, with the average mean (blue line), packages touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

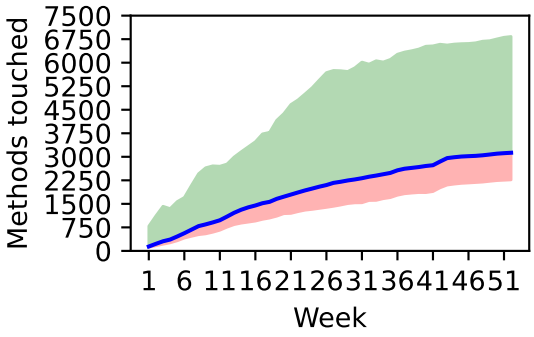


Sustained founder (n=34 & μ=746.15)

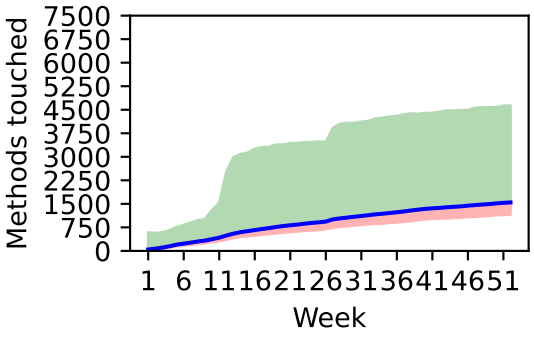


Sustained later joiner (n=203 & μ=408.05)

Figure 29: A time series for the all commits, excluding transient, with the average mean (blue line), classes touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.



Sustained founder (n=34 & μ=3129.0)



Sustained later joiner (n=203 & μ=1545.61)

Figure 30: A time series for the all commits, excluding transient, with the average mean (blue line), methods touched (y-axis) against the number of week (x-axis) for two (2) developer categories. Green is the standard deviation above the mean, and pink is the standard deviation below the mean.

1.6 Normality and homoscedasticity - For all components touched for all commits

Certain statistical methods require normality of each sample or the equality of variance. A Shapiro-Wilk Test evaluates whether a sample distribution significantly differs from the normal distribution. A Levene's test evaluates if the independent samples have equal variance.

Table 2: Diagnostic P-Value Thresholds Required for Statistical Method Selection

Statistical Method	**Shapiro-Wilk A (P)**	**Shapiro-Wilk B (P)**	**Levene's Test (P)**
Student's t-test	≥ 0.05 (Normal)	≥ 0.05 (Normal)	≥ 0.05 (Equal Variance)
Standard ANOVA	≥ 0.05 (Normal)	≥ 0.05 (Normal)	≥ 0.05 (Equal Variance)
Welch's t-test	≥ 0.05 (Normal)	≥ 0.05 (Normal)	< 0.05 (Unequal Variance)
Mann-Whitney U test	< 0.05 (Skewed)	< 0.05 (Skewed)	≥ 0.05 (Equal Variance)
Brunner-Münzel test	< 0.05 (Skewed)	< 0.05 (Skewed)	< 0.05 (Unequal Variance)
ART Anova test	< 0.05 (Skewed)	< 0.05 (Skewed)	< 0.05 (Unequal Variance)

Columns below:

- **Component** - Packages, classes or methods touched.
- **First *n* commits** - A sample of the first x number of commits for developers.
- **Contributor Category** Transient, Moderate or Sustained. "vs." equals versus.
- **Sample A** μ - The mean average of the first sample A. In brackets the number of developers for sample A.
- **Sample B** μ - The mean average of the second sample B. In brackets the number of developers for sample B.
- **Sample A Normality** P - Shapiro normality P value of sample A.
- **Sample B Normality** P - Shapiro normality P value of sample B.
- **Variance Equality** P - The Levene variance equality p-value between sample A and B.
- **Statistical Model** - A recommended statistical model of Welch t test; Mann-Whitney; Brunner Münzel
- **Model** P - The P value of the recommended test Welch t test; Mann-Whitney; Brunner Münzel

Table 3: A Shapiro-Wilk test of different categories of founder versus different categories of late joiner sample normality and a Levene's variance test between samples

Component	First *n* commits	Contributory Category	Founder μ	Joiner μ	Founder Normality P	Joiner Normality P	Variance Equality P	Statistical Models	Model P
Packages	1	All	0.44 (137)	0.10 (8303)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0001
Packages	1	Moderate	0.80 (40)	0.51 (580)	<0.001	<0.001	0.4760	Standard ANOVA, Mann-Whitney	0.2044
Packages	1	Sustained	0.38 (34)	0.65 (203)	<0.001	<0.001	0.5503	Standard ANOVA, Mann-Whitney	0.9673
Packages	5	Moderate & sustained	3.36 (74)	3.85 (783)	<0.001	<0.001	0.1596	Standard ANOVA, Mann-Whitney	0.7459
Packages	5	Moderate	3.52 (40)	3.72 (580)	<0.001	<0.001	0.3712	Standard ANOVA, Mann-Whitney	0.6567
Packages	5	Sustained	3.18 (34)	4.22 (203)	<0.001	<0.001	0.2932	Standard ANOVA, Mann-Whitney	0.8231
Packages	10	Moderate & sustained	5.07 (74)	6.05 (783)	<0.001	<0.001	0.0740	Standard ANOVA, Mann-Whitney	0.8498
Packages	10	Moderate	5.15 (40)	6.09 (580)	<0.001	<0.001	0.1245	Standard ANOVA, Mann-Whitney	0.9925
Packages	10	Sustained	4.97 (34)	5.94 (203)	<0.001	<0.001	0.4395	Standard ANOVA, Mann-Whitney	0.6772
Packages	20	Sustained	7.18 (34)	9.62 (203)	<0.001	<0.001	0.2278	Standard ANOVA, Mann-Whitney	0.4616
Classes	1	All	4.01 (137)	0.64 (8303)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0000
Classes	1	Moderate	7.47 (40)	2.37 (580)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.5349
Classes	1	Sustained	2.85 (34)	4.81 (203)	<0.001	<0.001	0.7276	Standard ANOVA, Mann-Whitney	0.2850
Classes	5	Moderate & sustained	39.05 (74)	24.11 (783)	<0.001	<0.001	0.0066	ANT ANOVA, Brunner-Münzel	0.1101
Classes	5	Moderate	41.38 (40)	22.04 (580)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.1057
Classes	5	Sustained	36.32 (34)	30.01 (203)	<0.001	<0.001	0.6508	Standard ANOVA, Mann-Whitney	0.5895
Classes	10	Moderate & sustained	64.81 (74)	43.51 (783)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0840
Classes	10	Moderate	61.60 (40)	41.95 (580)	<0.001	<0.001	0.0046	ANT ANOVA, Brunner-Münzel	0.1665
Classes	10	Sustained	68.59 (34)	47.98 (203)	<0.001	<0.001	0.1446	Standard ANOVA, Mann-Whitney	0.2478
Classes	20	Sustained	107.50 (34)	81.91 (203)	<0.001	<0.001	0.0906	Standard ANOVA, Mann-Whitney	0.2857
Methods	1	All	12.52 (137)	2.06 (8303)	<0.001	<0.001	0.0024	ANT ANOVA, Brunner-Münzel	0.0000
Methods	1	Moderate	24.27 (40)	7.22 (580)	<0.001	<0.001	0.0025	ANT ANOVA, Brunner-Münzel	0.5241
Methods	1	Sustained	10.00 (34)	23.40 (203)	<0.001	<0.001	0.7410	Standard ANOVA, Mann-Whitney	0.3092
Methods	5	Moderate & sustained	164.59 (74)	86.68 (783)	<0.001	<0.001	0.0072	ANT ANOVA, Brunner-Münzel	0.1274
Methods	5	Moderate	186.75 (40)	76.85 (580)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0851
Methods	5	Sustained	138.53 (34)	114.77 (203)	<0.001	<0.001	0.7156	Standard ANOVA, Mann-Whitney	0.7890
Methods	10	Moderate & sustained	263.26 (74)	154.41 (783)	<0.001	<0.001	0.0018	ANT ANOVA, Brunner-Münzel	0.0591
Methods	10	Moderate	271.43 (40)	145.21 (580)	<0.001	<0.001	0.0017	ANT ANOVA, Brunner-Münzel	0.1449
Methods	10	Sustained	253.65 (34)	180.70 (203)	<0.001	<0.001	0.3267	Standard ANOVA, Mann-Whitney	0.2291
Methods	20	Sustained	401.24 (34)	295.10 (203)	<0.001	<0.001	0.1120	Standard ANOVA, Mann-Whitney	0.5131

Table 4: A Shapiro-Wilk test of sample A category versus sample B category sample normality and a Levene's variance test between samples

Component	First *n* commits	Contributory Category	Sample A μ	Sample B μ	Sample A Normality P	Sample B Normality P	Variance Equality P	Statistical Models	Model P
Packages	1	Transient vs. moderate	0.05 (7583)	0.53 (620)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0000
Packages	1	Transient vs. sustained	0.05 (7583)	0.61 (237)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0000
Packages	1	Transient vs. moderate & sustained	0.05 (7583)	0.55 (857)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0000
Packages	1	Moderate vs. sustained	0.53 (620)	0.61 (237)	<0.001	<0.001	0.6750	Standard ANOVA, Mann-Whitney	0.5695
Packages	5	Moderate vs. sustained	3.71 (620)	4.07 (237)	<0.001	<0.001	0.8937	Standard ANOVA, Mann-Whitney	0.2931
Packages	10	Moderate vs. sustained	6.03 (620)	5.80 (237)	<0.001	<0.001	0.2223	Standard ANOVA, Mann-Whitney	0.9068
Classes	1	Transient vs. moderate	0.41 (7583)	2.70 (620)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0000
Classes	1	Transient vs. sustained	0.41 (7583)	4.53 (237)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0000
Classes	1	Transient vs. moderate & sustained	0.41 (7583)	3.21 (857)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0000
Classes	1	Moderate vs. sustained	2.70 (620)	4.53 (237)	<0.001	<0.001	0.1757	Standard ANOVA, Mann-Whitney	0.0871
Classes	5	Moderate vs. sustained	23.29 (620)	30.92 (237)	<0.001	<0.001	0.0129	ANT ANOVA, Brunner-Münzel	0.8160
Classes	10	Moderate vs. sustained	43.22 (620)	50.94 (237)	<0.001	<0.001	0.0197	ANT ANOVA, Brunner-Münzel	0.6548
Methods	1	Transient vs. moderate	1.12 (7583)	8.32 (620)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0000
Methods	1	Transient vs. sustained	1.12 (7583)	21.48 (237)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0000
Methods	1	Transient vs. moderate & sustained	1.12 (7583)	11.96 (857)	<0.001	<0.001	<0.001	ANT ANOVA, Brunner-Münzel	0.0000
Methods	1	Moderate vs. sustained	8.32 (620)	21.48 (237)	<0.001	<0.001	0.1457	Standard ANOVA, Mann-Whitney	0.0791
Methods	5	Moderate vs. sustained	83.94 (620)	118.18 (237)	<0.001	<0.001	0.0495	ANT ANOVA, Brunner-Münzel	0.9920
Methods	10	Moderate vs. sustained	153.35 (620)	191.16 (237)	<0.001	<0.001	0.0476	ANT ANOVA, Brunner-Münzel	0.9384

1.7 Statistical significance test - For all components touched for all commits

A Welch t-test was employed to determine if the mean contributions of Group A differ significantly from Group B. This test is appropriate for continuous data and is robust to unequal variances.

Complementarily, the Mann-Whitney U (MWU) test was used to assess stochastic dominance by ranking data points. This non-parametric approach does not require a normal distribution and is highly resistant to outliers.

The Brunner-Münzel test was employed as a robust non-parametric method to determine if the relative treatment effects of Group A differ significantly from Group B. This test is highly appropriate for skewed count data and is completely robust to unequal variances and large group size imbalances. Complementarily, Cliff's δ and the Probability of Superiority (PS) are provided as robust non-parametric effect sizes to quantify the directional dominance magnitude between the samples.

Columns below:

- **Component** - Packages, classes or method touched.
- **First *n* commits** - A sample of the first x number of commits for developers.
- **Contributor Category** Transient, Moderate or Sustained. "vs." equals versus.
- **Sample A** μ - The mean average of the first sample A. In brackets the number of developers for sample A.
- **Sample B** μ - The mean average of the second sample B. In brackets the number of developers for sample b.
- **Welch Statistic** - This is the magnitude of the change, greater than 2.0 generally indicates a significant change.
- **Welch** P - A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **MWU** P - The Mann-Whitney p-value. A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **BM** P - The Brunner-Münzel asymptotic p-value. A $p < 0.05$ indicates a significant difference in relative effects.
- **Brunner-Münzel W** - The robust test statistic. Large absolute values indicate a significant structural shift.
- **Cliff's** δ - Non-parametric effect size bound between -1 and +1, evaluating directional dominance.
- **Prob. of Superiority** (PS) - The empirical probability (0 to 1) that a randomly selected developer from Sample A outranked a developer from Sample B.

Table 5: Statistical significance tests of different categories of founder versus different categories of late joiner touches of different granularities of component (package, class, method) after 1, 5, 10 and 20 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

Component	First *n* commits	Contributory Category	Founder μ	Joiner μ	Welch Statistic	Welch P	MWU P	BM P	Brunner-Münzel W	Cliff's δ	Prob. Sup. (PS)
Packages	1	All	0.44 (137)	0.10 (8303)	2.76	0.0067	0.0000	0.0001	-4.11	0.130	0.565
Packages	1	Moderate	0.80 (40)	0.51 (580)	0.77	0.4477	0.2044	0.2833	-1.09	0.076	0.538
Packages	1	Sustained	0.38 (34)	0.65 (203)	-1.06	0.2921	0.9673	0.9662	-0.04	0.003	0.502
Packages	5	Moderate & sustained	3.36 (74)	3.85 (783)	-0.99	0.3232	0.7459	0.7243	-0.35	0.022	0.511
Packages	5	Moderate	3.52 (40)	3.72 (580)	-0.27	0.7874	0.6567	0.6232	-0.49	0.041	0.520
Packages	5	Sustained	3.18 (34)	4.22 (203)	-1.47	0.1459	0.8231	0.8181	0.23	-0.024	0.488
Packages	10	Moderate & sustained	5.07 (74)	6.05 (783)	-1.33	0.1855	0.8498	0.8336	0.21	-0.013	0.493
Packages	10	Moderate	5.15 (40)	6.09 (580)	-0.95	0.3462	0.9925	0.9911	-0.01	0.001	0.500
Packages	10	Sustained	4.97 (34)	5.94 (203)	-0.83	0.4084	0.6772	0.6707	0.43	-0.044	0.478
Packages	20	Sustained	7.18 (34)	9.62 (203)	-1.41	0.1628	0.4616	0.4391	0.78	-0.078	0.461
Classes	1	All	4.01 (137)	0.64 (8303)	2.92	0.0042	0.0000	0.0000	-7.17	0.309	0.654
Classes	1	Moderate	7.47 (40)	2.37 (580)	1.43	0.1603	0.4989	0.5349	-0.63	0.057	0.528
Classes	1	Sustained	2.85 (34)	4.81 (203)	-0.76	0.4494	0.2850	0.3160	-1.01	0.096	0.548
Classes	5	Moderate & sustained	39.05 (74)	24.11 (783)	1.77	0.0799	0.0872	0.1101	-1.61	0.120	0.560
Classes	5	Moderate	41.38 (40)	22.04 (580)	1.70	0.0969	0.0626	0.1057	-1.65	0.176	0.588
Classes	5	Sustained	36.32 (34)	30.01 (203)	0.48	0.6350	0.5895	0.5843	-0.55	0.058	0.529
Classes	10	Moderate & sustained	64.81 (74)	43.51 (783)	2.07	0.0419	0.0569	0.0840	-1.75	0.134	0.567
Classes	10	Moderate	61.60 (40)	41.95 (580)	1.40	0.1679	0.1180	0.1665	-1.41	0.148	0.574
Classes	10	Sustained	68.59 (34)	47.98 (203)	1.30	0.2003	0.2478	0.2786	-1.10	0.124	0.562
Classes	20	Sustained	107.50 (34)	81.91 (203)	1.34	0.1863	0.2857	0.3224	-1.00	0.115	0.557
Methods	1	All	12.52 (137)	2.06 (8303)	2.48	0.0144	0.0000	0.0000	-7.17	0.308	0.654
Methods	1	Moderate	24.27 (40)	7.22 (580)	1.32	0.1957	0.4881	0.5241	-0.64	0.058	0.529
Methods	1	Sustained	10.00 (34)	23.40 (203)	-0.77	0.4417	0.3092	0.3379	-0.97	0.092	0.546
Methods	5	Moderate & sustained	164.59 (74)	86.68 (783)	1.56	0.1218	0.0995	0.1274	-1.54	0.116	0.558
Methods	5	Moderate	186.75 (40)	76.85 (580)	1.46	0.1516	0.0499	0.0851	-1.76	0.185	0.593
Methods	5	Sustained	138.53 (34)	114.77 (203)	0.36	0.7211	0.7890	0.7946	-0.26	0.029	0.514
Methods	10	Moderate & sustained	263.26 (74)	154.41 (783)	1.76	0.0826	0.0358	0.0591	-1.91	0.148	0.574
Methods	10	Moderate	271.43 (40)	145.21 (580)	1.28	0.2063	0.0973	0.1449	-1.49	0.157	0.578
Methods	10	Sustained	253.65 (34)	180.70 (203)	1.00	0.3240	0.2291	0.2666	-1.13	0.129	0.565
Methods	20	Sustained	401.24 (34)	295.10 (203)	1.16	0.2533	0.5131	0.5561	-0.59	0.070	0.535

Table 6: Statistical significance tests of sample A category versus sample B category touches of different granularities of component (package, class, method) after 1, 5 and 10 commits for contributors. The significance has been calculated for the following statistical tests: Welch t-test, Mann-Whitney and Brunner Münzel.

Component	First *n* commits	Contributory Category	Sample A μ	Sample B μ	Welch Statistic	Welch P	MWU P	BM P	Brunner-Münzel W	Cliff's δ	Prob. Sup. (PS)
Packages	1	Transient vs. moderate	0.05 (7583)	0.53 (620)	-4.79	0.0000	0.0000	0.0000	9.47	-0.140	0.430
Packages	1	Transient vs. sustained	0.05 (7583)	0.61 (237)	-3.62	0.0004	0.0000	0.0000	6.27	-0.155	0.423
Packages	1	Transient vs. moderate & sustained	0.05 (7583)	0.55 (857)	-5.96	0.0000	0.0000	0.0000	11.33	-0.144	0.428
Packages	1	Moderate vs. sustained	0.53 (620)	0.61 (237)	-0.43	0.6686	0.5695	0.5775	0.56	-0.016	0.492
Packages	5	Moderate vs. sustained	3.71 (620)	4.07 (237)	-0.80	0.4217	0.2931	0.2820	1.08	-0.045	0.477
Packages	10	Moderate vs. sustained	6.03 (620)	5.80 (237)	0.40	0.6905	0.9068	0.9036	0.12	-0.005	0.497
Classes	1	Transient vs. moderate	0.41 (7583)	2.70 (620)	-6.31	0.0000	0.0000	0.0000	15.68	-0.316	0.342
Classes	1	Transient vs. sustained	0.41 (7583)	4.53 (237)	-2.10	0.0368	0.0000	0.0000	7.99	-0.249	0.376
Classes	1	Transient vs. moderate & sustained	0.41 (7583)	3.21 (857)	-4.64	0.0000	0.0000	0.0000	17.42	-0.297	0.351
Classes	1	Moderate vs. sustained	2.70 (620)	4.53 (237)	-0.91	0.3614	0.0871	0.0819	-1.74	0.066	0.533
Classes	5	Moderate vs. sustained	23.29 (620)	30.92 (237)	-1.69	0.0919	0.8148	0.8160	-0.23	0.010	0.505
Classes	10	Moderate vs. sustained	43.22 (620)	50.94 (237)	-1.51	0.1316	0.6500	0.6548	-0.45	0.020	0.510
Methods	1	Transient vs. moderate	1.12 (7583)	8.32 (620)	-5.16	0.0000	0.0000	0.0000	15.64	-0.315	0.342
Methods	1	Transient vs. sustained	1.12 (7583)	21.48 (237)	-1.44	0.1522	0.0000	0.0000	7.99	-0.249	0.375
Methods	1	Transient vs. moderate & sustained	1.12 (7583)	11.96 (857)	-2.68	0.0076	0.0000	0.0000	17.39	-0.297	0.351
Methods	1	Moderate vs. sustained	8.32 (620)	21.48 (237)	-0.92	0.3564	0.0791	0.0733	-1.80	0.068	0.534
Methods	5	Moderate vs. sustained	83.94 (620)	118.18 (237)	-1.48	0.1389	0.9921	0.9920	-0.01	0.000	0.500
Methods	10	Moderate vs. sustained	153.35 (620)	191.16 (237)	-1.49	0.1366	0.9372	0.9384	0.08	-0.003	0.498

Statistical Models

Welch's t-test To specifically compare the two primary roles (**Sample A** vs. **Sample B**) without assuming equal variances or sample sizes, Welch's t-test is employed. The t-statistic is defined as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \tag{1}$$

Where:

- $\bar{X}_1, \bar{X}_2$ are the sample means of Sample A and Sample B.
- s_1^2, s_2^2 are the sample variances.
- N_1, N_2 are the sample sizes.

The degrees of freedom (ν) for this test are calculated using the Welch-Satterthwaite equation:

$$\nu \approx \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{(s_1^2/N_1)^2}{N_1 - 1} + \frac{(s_2^2/N_2)^2}{N_2 - 1}} \tag{2}$$

Mann-Whitney U Test To evaluate the differences in the stochastic dominance of architectural component exploration between primary roles (**Sample A** vs. **Sample B**) without assuming a normal distribution, the non-parametric Mann-Whitney U test (Wilcoxon rank-sum test) is employed.

The data from both independent groups are combined and ranked globally from lowest (1) to highest ($N_1 + N_2$), resolving ties by assigning the average rank. The U statistic for Sample A (U_1) is calculated as:

$$U_1 = R_1 - \frac{N_1(N_1 + 1)}{2} \tag{3}$$

Where:

- N_1 is the sample size of Sample A.
- R_1 is the sum of the ranks assigned to Sample A.
- The total number of pairwise comparisons (maximum possible pairs) is defined by $N_1 \times N_2$.

Normal Approximation for Large Samples Given the substantial sample sizes present in repository participant metrics, the distribution of the U statistic rapidly approaches a normal distribution. The asymptotic significance is determined using a standard normal Z-score approximation:

$$Z = \frac{U_1 - \mu_U}{\sigma_U} \tag{4}$$

Where the expected mean (μ_U) and standard deviation (σ_U) under the null hypothesis of identical distributions are defined as:

$$\mu_U = \frac{N_1 N_2}{2}, \quad \sigma_U = \sqrt{\frac{N_1 N_2 (N_1 + N_2 + 1)}{12}} \tag{5}$$

A calculated p-value lower than the alpha threshold ($p < 0.05$) indicates a statistically significant difference in the rank-order distributions of the two groups, demonstrating that one developer role systematically outranks the other in codebase exploration.

Brunner-Münzel Test To test for differences in the stochastic distributions of primary roles (**Sample A** vs. **Sample B**) without assuming normal distribution or variance homogeneity, the Brunner-Münzel test is applied. The test statistic W is defined as:

$$W = \sqrt{\frac{N_1 N_2}{N_1 + N_2}} \cdot \frac{\hat{p} - \frac{1}{2}}{\sqrt{\hat{\sigma}^2}} \tag{6}$$

Where $\hat{p}$ represents the relative treatment effect (Probability of Superiority), reflecting the probability that a random observation from Sample A is larger than one from Sample B plus half the probability that they are equal.

Cliff's δ Effect Size To complement hypothesis significance testing with a distribution-free magnitude metric, Cliff's δ is tracked to express the degree of dominance of one sample over the other:

$$\delta = \frac{\sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \text{sign}(X_{1i} - X_{2j})}{N_1 N_2} \tag{7}$$

Where the sign function yields $+1$ when an element of Sample A is larger than Sample B, 0 if equal, and -1 if smaller. Effect sizes are interpreted using standard empirical thresholds ($|\delta| < 0.147$ is negligible, ≥ 0.147 is small, ≥ 0.330 is medium, and ≥ 0.474 is large).

1.8 Generated standard anova - For all components touched for all commits

An Anova table for all components touched on average.

The Two-Way ANOVA evaluates the influence of two independent categorical variables—*Role* (Founder vs. Late Joiner) and *Tenure* (Moderate vs. Sustained)—on the number of unique architectural components touched. The analysis excludes and includes Transient developers to isolate the behaviors of contributors with established project engagement.

Explanation of Table Columns:

- **Sum Sq (Sum of Squares):** Represents the total variation explained by that specific factor. A higher Sum Sq relative to the Residual indicates that the factor has a strong influence on the outcome.
- **df (Degrees of Freedom):** The number of independent values that can vary in the calculation. For categorical variables, this is typically $N - 1$ levels.
- **F (F-statistic):** The ratio of the variance explained by the factor to the variance unexplained (Residual). A larger F-value suggests the factor is more likely to be significant.
- **p-Value ($PR(> F)$):** The probability that the observed differences occurred by chance. Using a significance threshold of $\alpha = 0.05$, values above this indicate no statistically significant effect.
- **ETA Sq** The effect size, quantifies the proportion of the total variance in your dependent variable.

Explanation of Table Rows:

- **Intercept:** Represents the baseline value of the dependent variable when all categorical factors are at their reference levels.
- **Role(F/L):** Tests the *Main Role Effect* of being a Founder versus a Late Joiner. The p-Value above 0.05 indicates that the timing of project entry does not significantly change the breadth of early-stage component exploration.
- **Category(M/S):** Tests the *Main Category Effect* of the eventual contributor catregory (Transient vs Moderate vs. Sustained).
- **Interaction:** Tests the *Interaction Effect*. It evaluates if the effect of being a Founder is different for the different categories.
- **Residual:** Represents the "noise" or variance within the groups that cannot be explained by Role or Tenure.

1.8.1 Generated standard anova - For all components touched for a all commits excluding tranient

Table 7: Anova results for all components, number of commits and 8440 developers excluding transient

Component	Number of First Commits	Factor	Sum Squared	df	F-Stat	*P* Value	ETA Sq
packages	1	Role (F/LJ)	3.14	1	3.31	0.0688	0.0004
packages	1	Category (T/M/S)	7.86	2	4.15	0.0158	0.0010
packages	1	Interaction	5.65	2	2.98	0.0509	0.0007
packages	1	Residual	7995.45	8434	-	-	
packages	5	Role (F/LJ)	1.43	1	0.05	0.8250	0.0001
packages	5	Category (M/S)	2.23	1	0.08	0.7826	0.0001
packages	5	Interaction	11.68	1	0.40	0.5279	0.0005
packages	5	Residual	24984.13	853	-	-	
packages	10	Role (F/LJ)	33.04	1	0.59	0.4443	0.0007
packages	10	Category (M/S)	0.59	1	0.01	0.9185	0.0000
packages	10	Interaction	0.01	1	0.00	0.9889	0.0000
packages	10	Residual	48127.58	853	-	-	
classes	1	Role (F/LJ)	973.61	1	24.93	0.0000	0.0029
classes	1	Category (T/M/S)	683.32	2	8.75	0.0002	0.0020
classes	1	Interaction	815.33	2	10.44	0.0000	0.0024
classes	1	Residual	329384.23	8434	-	-	
classes	5	Role (F/LJ)	13981.95	1	7.35	0.0069	0.0082
classes	5	Category (M/S)	468.97	1	0.25	0.6198	0.0003
classes	5	Interaction	2774.71	1	1.46	0.2276	0.0016
classes	5	Residual	1623543.63	853	-	-	
classes	10	Role (F/LJ)	14448.45	1	5.02	0.0254	0.0055
classes	10	Category (M/S)	897.52	1	0.31	0.5769	0.0003
classes	10	Interaction	15.03	1	0.01	0.9424	0.0000
classes	10	Residual	2457107.31	853	-	-	
methods	1	Role (F/LJ)	10883.40	1	6.87	0.0088	0.0008
methods	1	Category (T/M/S)	8092.96	2	2.56	0.0777	0.0006
methods	1	Interaction	15271.02	2	4.82	0.0081	0.0011
methods	1	Residual	13353418.84	8434	-	-	
methods	5	Role (F/LJ)	451979.70	1	8.27	0.0041	0.0093
methods	5	Category (M/S)	42733.87	1	0.78	0.3769	0.0009
methods	5	Interaction	121524.53	1	2.22	0.1364	0.0025
methods	5	Residual	46646591.43	853	-	-	
methods	10	Role (F/LJ)	596144.39	1	7.60	0.0060	0.0085
methods	10	Category (M/S)	5808.58	1	0.07	0.7856	0.0001
methods	10	Interaction	46476.28	1	0.59	0.4417	0.0007
methods	10	Residual	66927848.79	853	-	-	

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{8}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Late Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{9}$$

1.9 Tukey hsd - For all components touched for all commits

A Tukey Honestly Significant Difference table for all components touched on average.

Post-hoc Analysis (Tukey HSD): Following the ANOVA, a Tukey Honestly Significant Difference (HSD) test was performed. While ANOVA identifies if at least one group differs, Tukey HSD performs pairwise comparisons to identify specific differences between developer roles and tenure categories while controlling for Type I error rates across multiple comparisons.

Table 8: Significant Tukey HSD results for classes and number of commits, for 8440 developers

Component	Commits	Group A	Group B	Mean Diff	Adj P Value
classes	1	Founder Moderate	Founder Sustained	-4.62	0.0190
classes	1	Founder Moderate	Founder Transient	-5.05	0.0009
classes	1	Founder Moderate	Later Joiner Moderate	-5.10	0.0000
classes	1	Founder Moderate	Later Joiner Transient	-7.08	0.0000
classes	1	Later Joiner Moderate	Later Joiner Sustained	2.43	0.0000
classes	1	Later Joiner Moderate	Later Joiner Transient	-1.98	0.0000
classes	1	Later Joiner Sustained	Later Joiner Transient	-4.42	0.0000
classes	5	Founder Moderate	Later Joiner Moderate	-19.33	0.0345
classes	10	Founder Sustained	Later Joiner Moderate	-26.64	0.0258
methods	1	Founder Moderate	Later Joiner Transient	-23.19	0.0032
methods	1	Founder Transient	Later Joiner Sustained	16.99	0.0364
methods	1	Later Joiner Moderate	Later Joiner Sustained	16.18	0.0000
methods	1	Later Joiner Moderate	Later Joiner Transient	-6.14	0.0046
methods	1	Later Joiner Sustained	Later Joiner Transient	-22.32	0.0000
methods	5	Founder Moderate	Later Joiner Moderate	-109.90	0.0215
methods	10	Founder Moderate	Later Joiner Moderate	-126.22	0.0303

Table 9: Not Significant Tukey HSD results for classes and number of commits, for 8440 developers

Component	Commits	Group A	Group B	Mean Diff	Adj P Value
classes	1	Founder Moderate	Later Joiner Sustained	-2.67	0.1341
classes	1	Founder Sustained	Founder Transient	-0.42	0.9996
classes	1	Founder Sustained	Later Joiner Moderate	-0.48	0.9981
classes	1	Founder Sustained	Later Joiner Sustained	1.95	0.5397
classes	1	Founder Sustained	Later Joiner Transient	-2.46	0.1975
classes	1	Founder Transient	Later Joiner Moderate	-0.05	1.0000
classes	1	Founder Transient	Later Joiner Sustained	2.38	0.0880
classes	1	Founder Transient	Later Joiner Transient	-2.04	0.1031
classes	5	Founder Moderate	Founder Sustained	-5.05	0.9599
classes	5	Founder Moderate	Later Joiner Sustained	-11.37	0.4344
classes	5	Founder Sustained	Later Joiner Moderate	-14.28	0.2486
classes	5	Founder Sustained	Later Joiner Sustained	-6.31	0.8631
classes	5	Later Joiner Moderate	Later Joiner Sustained	7.96	0.1137
classes	10	Founder Moderate	Founder Sustained	6.99	0.9443
classes	10	Founder Moderate	Later Joiner Moderate	-19.65	0.1135
classes	10	Founder Moderate	Later Joiner Sustained	-13.62	0.4581
classes	10	Founder Sustained	Later Joiner Sustained	-20.61	0.1631
classes	10	Later Joiner Moderate	Later Joiner Sustained	6.03	0.5137
classes	20	Founder Sustained	Later Joiner Sustained	-25.59	0.1034
methods	1	Founder Moderate	Founder Sustained	-14.28	0.6396
methods	1	Founder Moderate	Founder Transient	-17.86	0.2283
methods	1	Founder Moderate	Later Joiner Moderate	-17.05	0.0921
methods	1	Founder Moderate	Later Joiner Sustained	-0.88	1.0000
methods	1	Founder Sustained	Founder Transient	-3.59	0.9983
methods	1	Founder Sustained	Later Joiner Moderate	-2.78	0.9988
methods	1	Founder Sustained	Later Joiner Sustained	13.40	0.4546
methods	1	Founder Sustained	Later Joiner Transient	-8.92	0.7829
methods	1	Founder Transient	Later Joiner Moderate	0.81	1.0000
methods	1	Founder Transient	Later Joiner Transient	-5.33	0.8974
methods	5	Founder Moderate	Founder Sustained	-48.22	0.8132
methods	5	Founder Moderate	Later Joiner Sustained	-71.98	0.2840
methods	5	Founder Sustained	Later Joiner Moderate	-61.68	0.4410
methods	5	Founder Sustained	Later Joiner Sustained	-23.76	0.9470
methods	5	Later Joiner Moderate	Later Joiner Sustained	37.92	0.1931
methods	10	Founder Moderate	Founder Sustained	-17.78	0.9930
methods	10	Founder Moderate	Later Joiner Sustained	-90.73	0.2408
methods	10	Founder Sustained	Later Joiner Moderate	-108.44	0.1257
methods	10	Founder Sustained	Later Joiner Sustained	-72.95	0.4963
methods	10	Later Joiner Moderate	Later Joiner Sustained	35.49	0.4058
methods	20	Founder Sustained	Later Joiner Sustained	-106.14	0.1554

Post-hoc Analysis: Tukey HSD

Following the Two-Way ANOVA, a Tukey Honestly Significant Difference (HSD) test is used to conduct pairwise comparisons. This test determines which specific group means differ significantly while maintaining a constant family-wise error rate.

The studentized range statistic (q) for comparing two group means is defined as:

$$q = \frac{\bar{Y}_{max} - \bar{Y}_{min}}{\sqrt{\frac{MS_{residual}}{n}}} \tag{10}$$

Where:

- $\bar{Y}_{max}$ and $\bar{Y}_{min}$ are the larger and smaller means of the two groups being compared.
- $MS_{residual}$ is the Mean Square Error derived from the ANOVA residual.
- n is the number of observations in each group (adjusted for unequal sample sizes via the harmonic mean).

A pair of groups is considered significantly different if the absolute difference between their means exceeds the Honestly Significant Difference (HSD):

$$HSD = q_{\alpha,k,df}\sqrt{\frac{MS_{residual}}{n}} \tag{11}$$

Where $q_{\alpha,k,df}$ is the critical value from the studentized range distribution based on the significance level ($\alpha = 0.05$), the number of groups (k), and the degrees of freedom (df) from the ANOVA residual.

1.10 Strategy logistic regression - For all components touched for all commits

To identify early indicators of long-term developer engagement, a Logistic Regression model was employed. The model treats the developer category (Sustained vs. Non-Sustained) as a binary dependent variable. The following metrics are reported:

- **Coefficient (β):** This value represents the change in the log-odds of a developer becoming 'Sustained' for every unit increase in the component count. A positive coefficient indicates that higher activity in that component increases the likelihood of long-term retention.
- **Odds Ratio (e^{β}):** Perhaps the most interpretable metric, the Odds Ratio quantifies the relative increase in the likelihood of being a Sustained developer. For example, an Odds Ratio of 1.05 implies a 5% increase in the odds of becoming sustained for each additional component touched.
- **Std. Err (Standard Error):** Measures the accuracy of the coefficient estimate. A lower standard error relative to the coefficient indicates a more precise estimate.
- **p-Value ($P > |z|$):** Indicates the statistical significance of the predictor. In this study, a p-Value < 0.05 suggests that the specific component count is a significant predictor of whether a developer will transition into a sustained role.
- **50% Threshold:** The threshold number of components to touch to have a 50
- **Moderate Met Threshold:** The moderate number of developers to have met the threshold, the percentage of developers in brackets.
- **Sustained Met Threshold:** The sustained number of developers to have met the threshold, the percentage of developers in brackets.

Interpretation of Intercept vs. Component Count:

- **Intercept:** Represents the baseline log-odds of the dependent variable (Sustained Status) when the independent variable (e.g., Package Count) is zero. In this study, the consistently negative intercepts indicate that the probability of a developer becoming 'Sustained' is statistically low without active engagement in the codebase.
- **Component Count (Slope):** Represents the effect size of architectural engagement. The positive coefficients and Odds Ratios greater than 1.0 demonstrate that every additional unique component touched by a developer significantly increases their likelihood of transitioning into a sustained contributor.

Table 10: Strategy Logistic Regression for all components against number of commits for 8440 developers excluding transient

Component	Number of First Commits	Type	Moderate μ	Sustained μ	Coefficient	Odds Ratio	Std. Err	p-Value	30% Threshold	Moderate Met Threshold	Sustained Met Threshold
packages	1	Intercept	0.40	3.99	-3.8260	0.0218	0.0752	0.0000	-	-	-
packages	1	Packages Count	0.40	3.99	0.2270	1.2548	0.0166	0.0000	**13.1243**	29 (0.4%)	17 (7.2%)
packages	5	Intercept	7.36	16.53	-1.3025	0.2719	0.0978	0.0000	-	-	-
packages	5	Packages Count	7.36	16.53	0.0324	1.0329	0.0055	0.0000	**14.0420**	116 (18.7%)	80 (33.8%)
packages	10	Intercept	9.68	23.99	-1.4121	0.2436	0.1019	0.0000	-	-	-
packages	10	Packages Count	9.68	23.99	0.0309	1.0313	0.0044	0.0000	**18.3064**	123 (19.8%)	90 (38.0%)
classes	1	Classes Count	2.82	36.49	0.0419	1.0428	0.0026	0.0000	**74.0370**	44 (0.5%)	28 (11.8%)
classes	5	Intercept	56.38	240.08	-1.8719	0.1538	0.1205	0.0000	-	-	-
classes	5	Classes Count	56.38	240.08	0.0092	1.0092	0.0009	0.0000	**111.4174**	86 (13.9%)	125 (52.7%)
classes	10	Intercept	79.55	407.66	-2.5977	0.0744	0.1560	0.0000	-	-	-
classes	10	Classes Count	79.55	407.66	0.0107	1.0108	0.0009	0.0000	**162.8939**	66 (10.6%)	160 (67.5%)
methods	1	Methods Count	9.44	133.10	0.0084	1.0084	0.0006	0.0000	**355.1713**	34 (0.4%)	16 (6.8%)
methods	5	Intercept	207.63	926.61	-1.6702	0.1882	0.1087	0.0000	-	-	-
methods	5	Methods Count	207.63	926.61	0.0019	1.0019	0.0002	0.0000	**438.0841**	67 (10.8%)	113 (47.7%)
methods	10	Intercept	289.62	1574.80	-2.2477	0.1056	0.1361	0.0000	-	-	-
methods	10	Methods Count	289.62	1574.80	0.0022	1.0022	0.0002	0.0000	**624.0901**	63 (10.2%)	150 (63.3%)

Predictive Identification Model

To determine if early-stage developer behavior can identify future sustained contributors, we employ a Logistic Regression model. The probability P of a developer becoming a **Sustained Contributor** is modeled using the log-odds (logit) function:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \epsilon \tag{12}$$

Where:

- P is the probability that the developer's tenure is "Sustained".
- β_0 is the intercept.
- X_n represents early-stage metrics (e.g., methods or classes touched in the first 10, 20, or 50 commits).
- β_n are the coefficients representing the weight of each predictor.

The impact of each behavior is quantified using the **Odds Ratio (OR)**, calculated as:

$$OR = e^{\beta_n} \tag{13}$$

An $OR > 1$ indicates that an increase in the metric (e.g., touching more classes) increases the likelihood of long-term retention.

Model Evaluation The predictive power is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). An AUC of 0.5 represents random chance, while an AUC closer to 1.0 indicates high predictive accuracy.

2 Repository: 14

2.1 Time series sustained components touched by developer - 14 For developer commits and components touched.

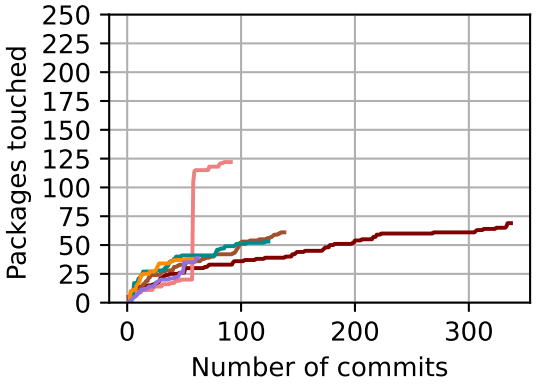
Table 11: Table of first 6 sustained founder developers in the Scatter Developer graphs.

Developer ID	Colour
2930	Maroon
2932	Sienna
2934	LightCoral
2935	DarkCyan
2946	DarkOrange
2959	MediumPurple

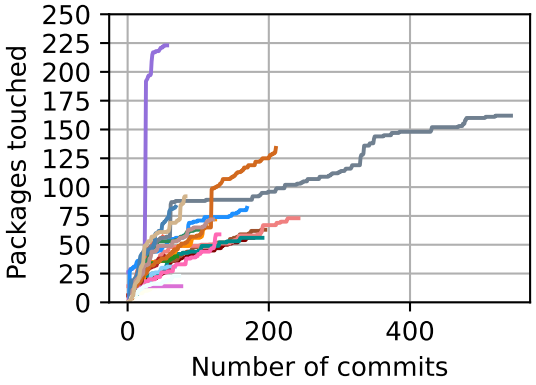
Table 12: Table of first 21 sustained later joiner developers in the Scatter Developer graphs.

Developer ID	Colour
1774	Maroon
2986	Sienna
3000	LightCoral
3020	DarkCyan
3037	DarkOrange
3050	MediumPurple
3071	LightSkyBlue
3072	HotPink
3141	ForestGreen
3143	FireBrick
3163	DodgerBlue
3172	Tomato
3178	SeaGreen
3216	SlateGray
3217	Ivory
3272	Orchid
3339	Chocolate
3354	SteelBlue
3456	RosyBrown
3473	Azure
3485	Tan

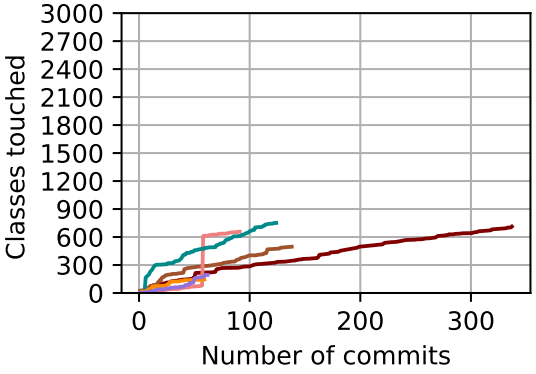
2.2 Time series sustained components touched by developer - 14 For developer day and components touched.



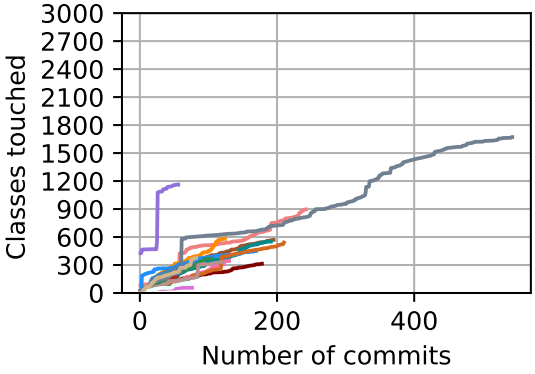
Packages for sustained founder



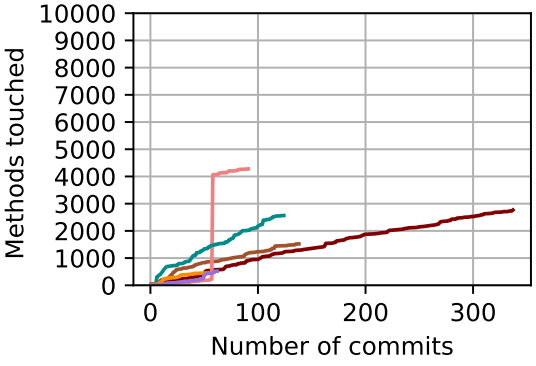
Packages for sustained later joiner



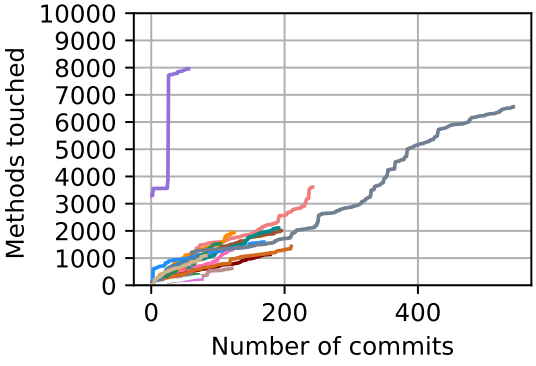
Classes for sustained founder



Classes for sustained later joiner

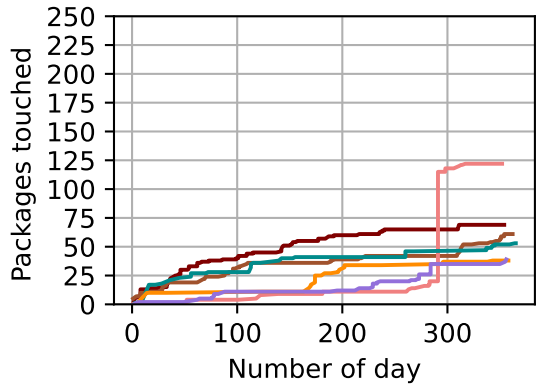


Methods for sustained founder

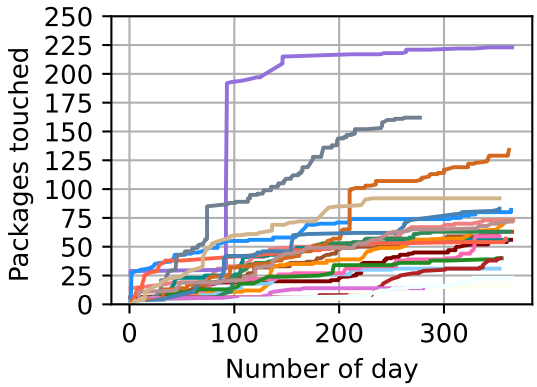


Methods for sustained later joiner

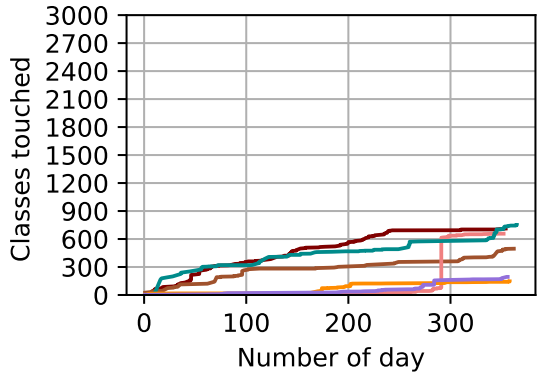
Figure 31: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder



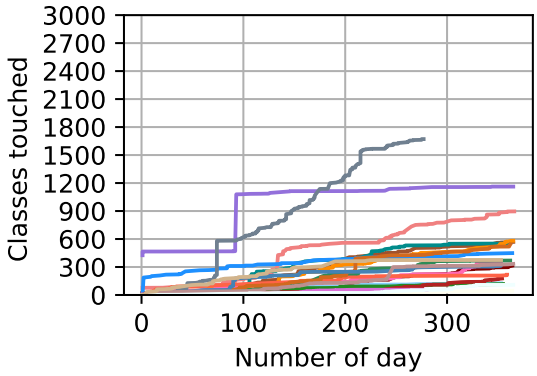
Packages for sustained founder



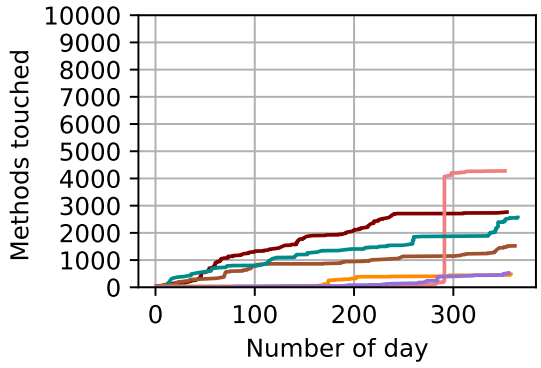
Packages for sustained later joiner



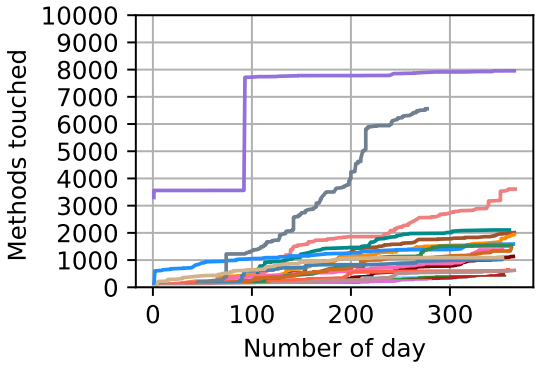
Classes for sustained founder



Classes for sustained later joiner



Methods for sustained founder



Methods for sustained later joiner

Figure 32: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 11. Sustained later joiner developers with colours are shown in Table 12.

3 Repository: 21

3.1 Time series sustained components touched by developer - 21 For developer commits and components touched.

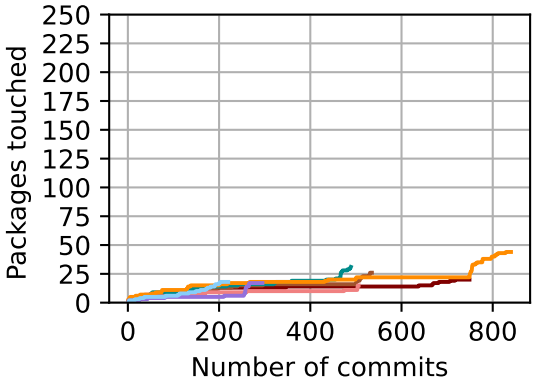
Table 13: Table of first 7 sustained founder developers in the Scatter Developer graphs.

Developer ID	Colour
6245	Maroon
6246	Sienna
6247	LightCoral
6249	DarkCyan
6250	DarkOrange
6251	MediumPurple
6257	LightSkyBlue

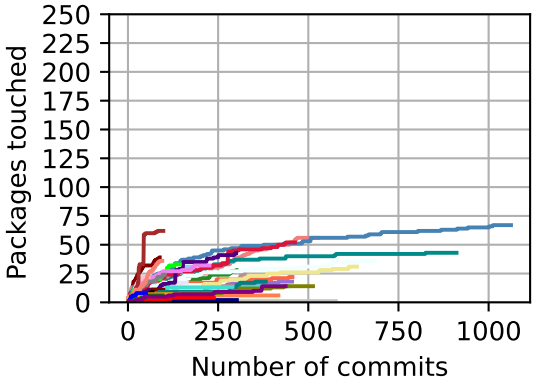
Table 14: Table of first 50 sustained later joiner developers in the Scatter Developer graphs.

Developer ID	Colour
199	Maroon
6252	Sienna
6253	LightCoral
6259	DarkCyan
6260	DarkOrange
6262	MediumPurple
6263	LightSkyBlue
6264	HotPink
6267	ForestGreen
6271	FireBrick
6272	DodgerBlue
6276	Tomato
6278	SeaGreen
6280	SlateGray
6282	Ivory
6284	Orchid
6288	Chocolate
6295	SteelBlue
6297	RosyBrown
6298	Azure
6302	Tan
6306	Plum
6309	Crimson
6312	Khaki
6313	Lavender
6314	Indigo
6315	Turquoise
6326	Beige
6329	Silver
6333	Gold
6338	Coral
6342	Lime
6343	Maroon
6349	Navy
6352	Teal
6355	Olive
6363	Violet
6368	Gray
6369	Pink
6370	Brown
6373	Black
6374	Yellow
6375	Magenta
6379	Cyan
6391	Salmon
6410	Orange
6411	Red
6417	Green
6418	Blue
6420	Purple

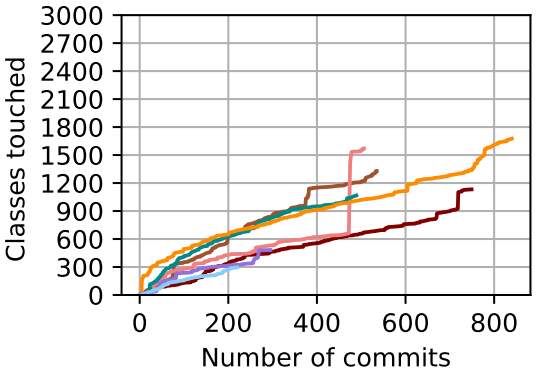
3.2 Time series sustained components touched by developer - 21 For developer day and components touched.



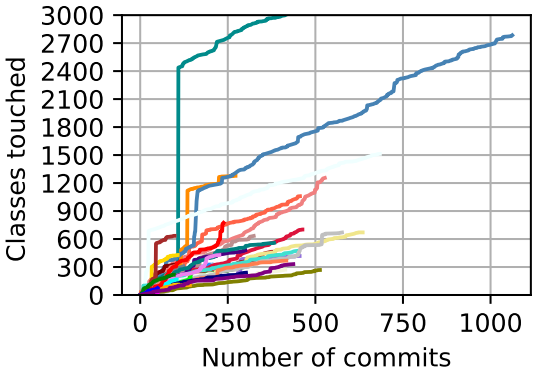
Packages for sustained founder



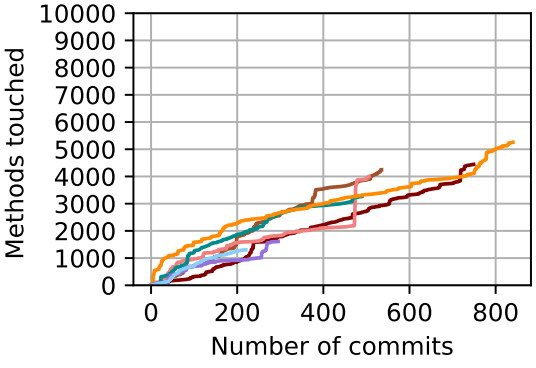
Packages for sustained later joiner



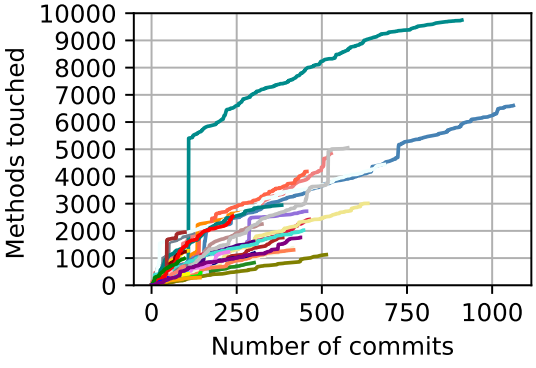
Classes for sustained founder



Classes for sustained later joiner

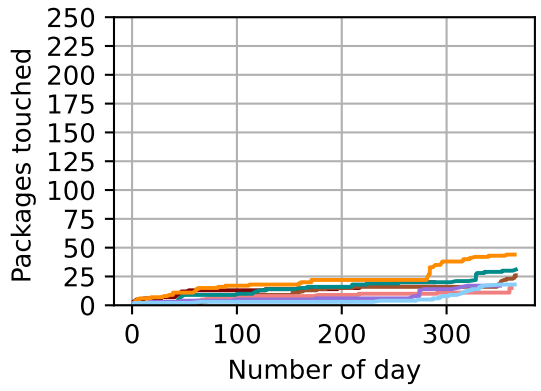


Methods for sustained founder

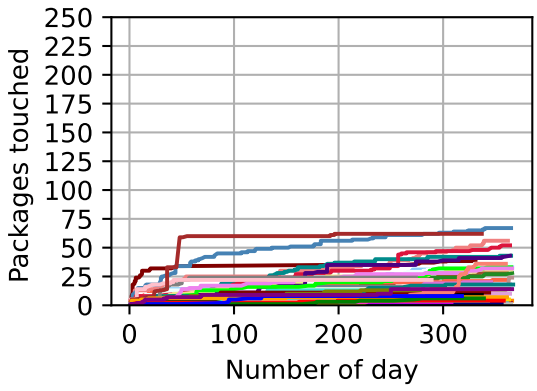


Methods for sustained later joiner

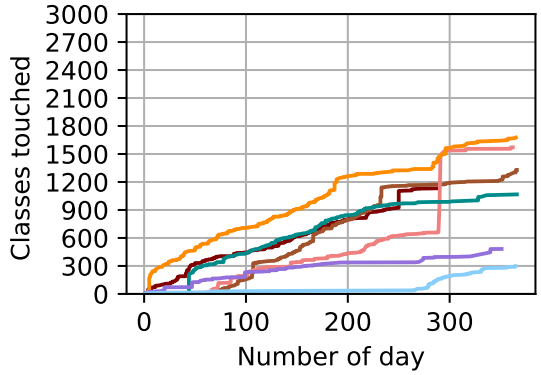
Figure 33: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder



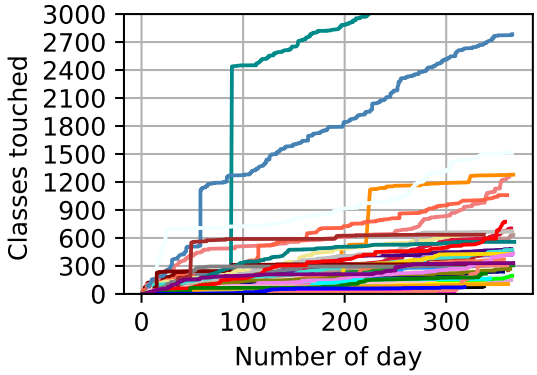
Packages for sustained founder



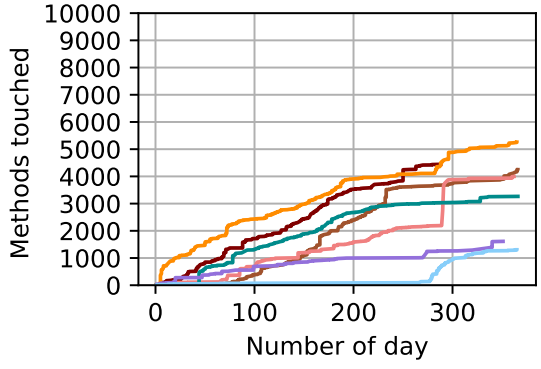
Packages for sustained later joiner



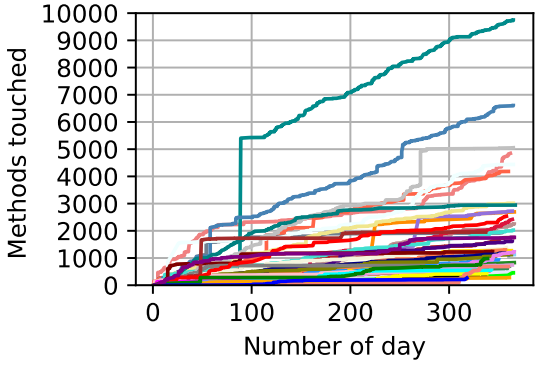
Classes for sustained founder



Classes for sustained later joiner



Methods for sustained founder



Methods for sustained later joiner

Figure 34: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 13. Sustained later joiner developers with colours are shown in Table 14.